

CCSE-F102 — Microsoft AZ-900: Azure Fundamentals

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Cloud Security Engineering

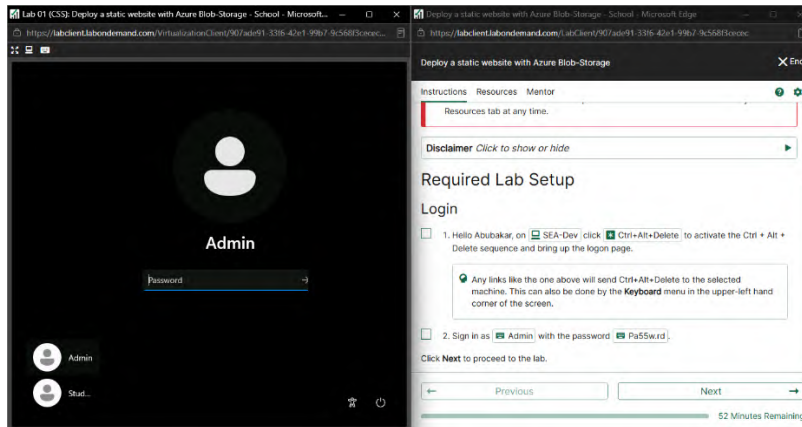
C11/26/CCSE/17867

**Microsoft AZ-900: Azure Fundamentals: Lab Assignment
(CCSE-F102)**

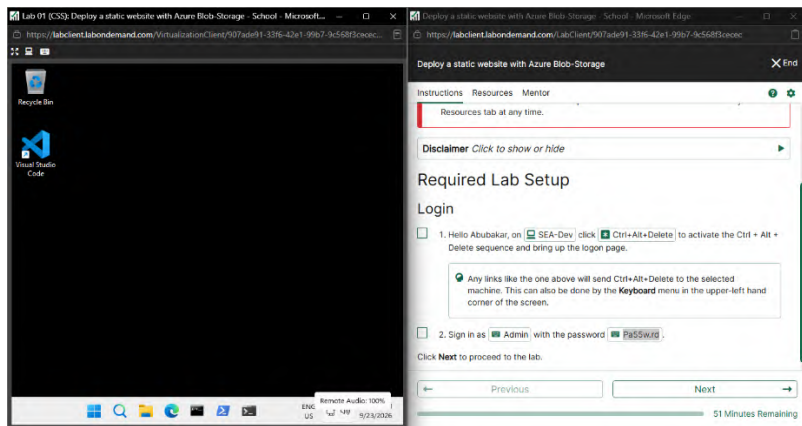
Mr. Aminu Idris, AMCPN

01/10/2026

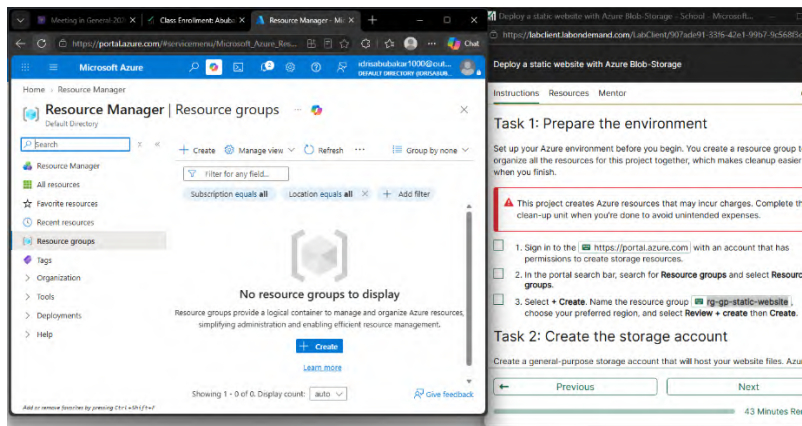
Deploy a static website with Azure Blob-Storage.



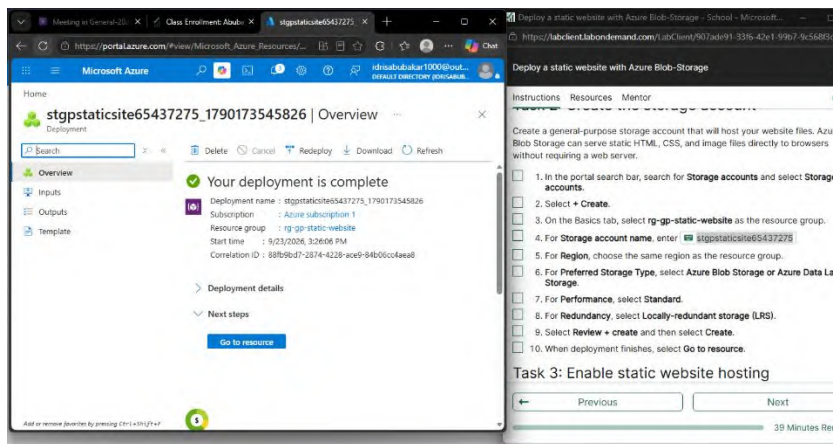
The image displays the initial login step of a virtual lab where the user is prompted to sign into the "Admin" account using the password specified in the adjacent setup instructions.



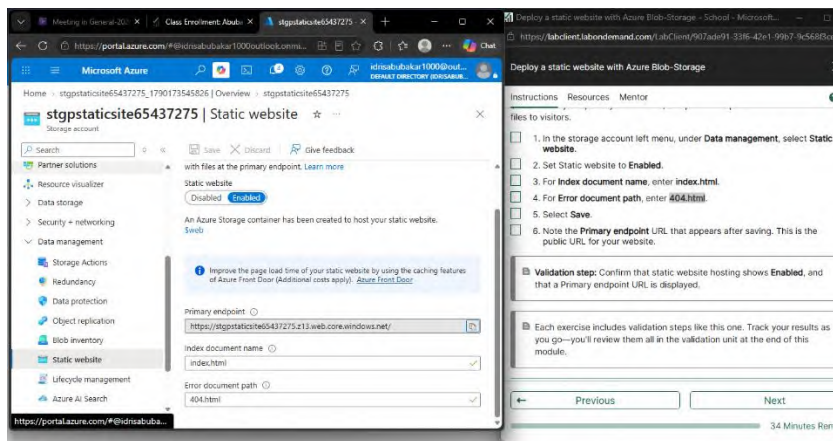
The user has successfully completed the required login steps from the lab instructions to access the Windows desktop environment in the virtual machine.



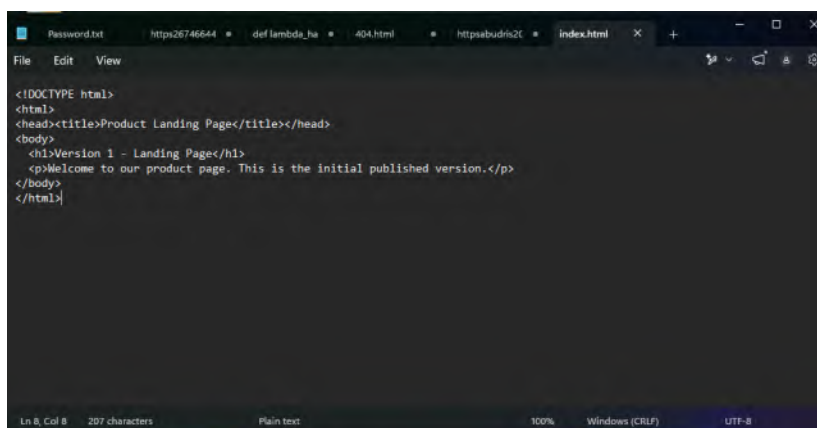
The user has logged into the Azure portal and navigated to the Resource groups section as part of the initial lab instructions to prepare the environment for a static website project.



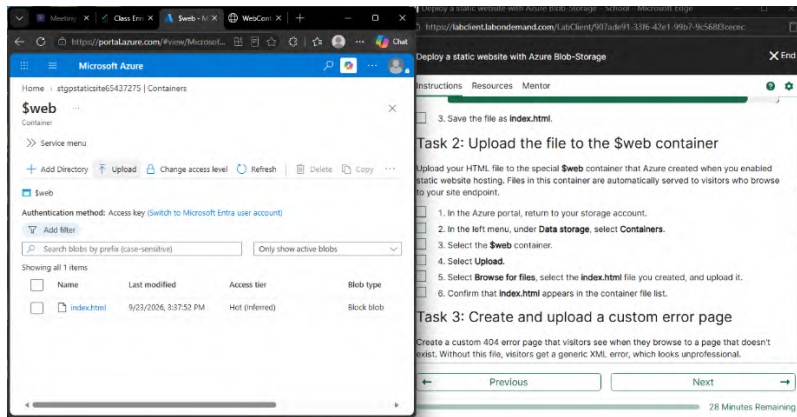
The user successfully completed the lab instructions to create and deploy a new Azure storage account named stgpstaticsite65437275 within the rg-gp-static-website resource group.



The user successfully enabled static website hosting on their Azure storage account by configuring the index and error document paths as specified in the lab instructions.



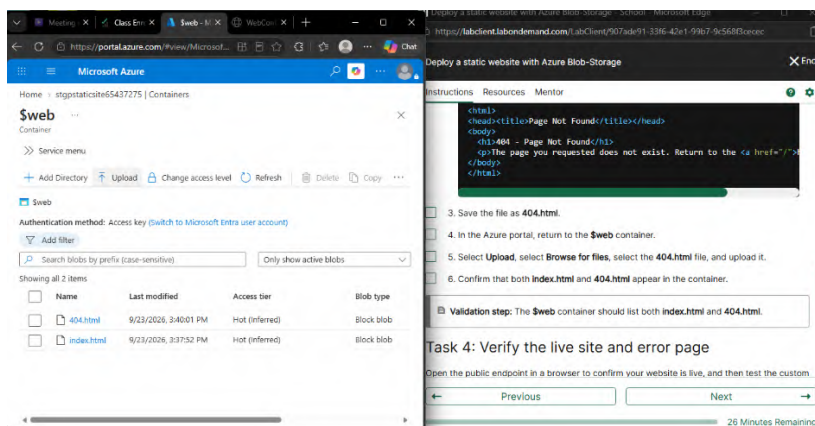
The user is viewing or editing the basic HTML source code for an initial product landing page within a text editor tab named index.html.



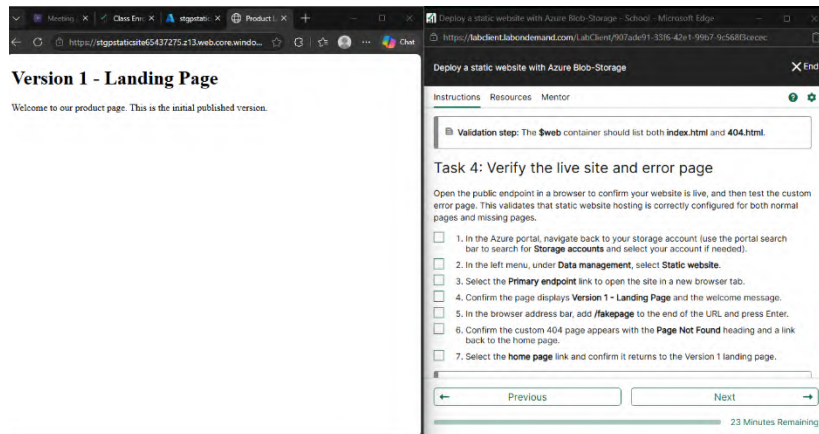
The user has successfully followed the lab instructions to upload their index.html file into the \$web container of their Azure storage account.



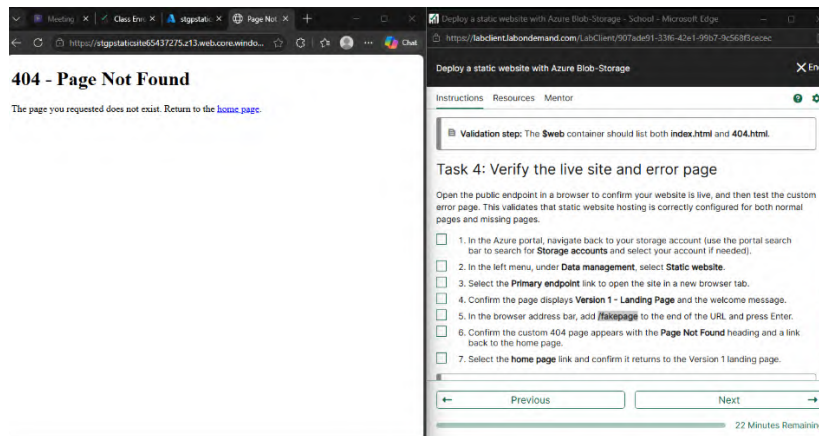
The user is writing the HTML source code for a custom "Page Not Found" error document within a text editor file named 404.html.



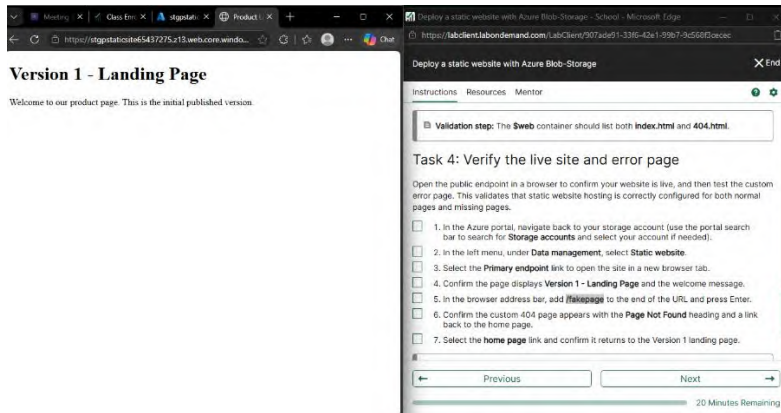
The user has successfully uploaded their custom 404.html error page into the \$web storage container alongside the existing index.html file, completing the validation step to ensure both files are present for their static website deployment.



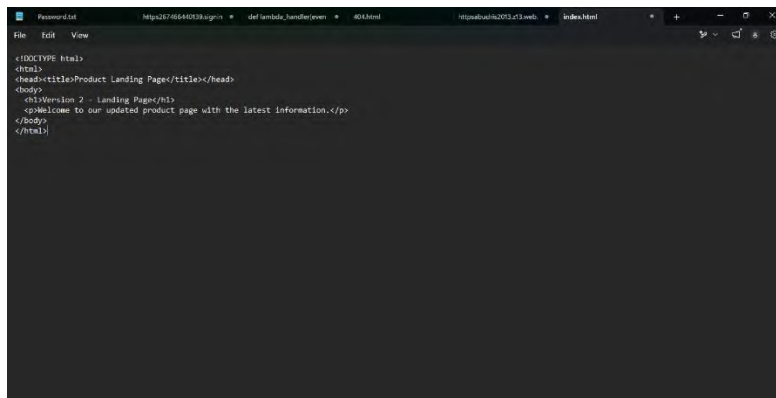
The user has successfully opened their primary endpoint URL in a web browser to verify that the static website is live and correctly displaying the uploaded landing page content as instructed in Task 4.



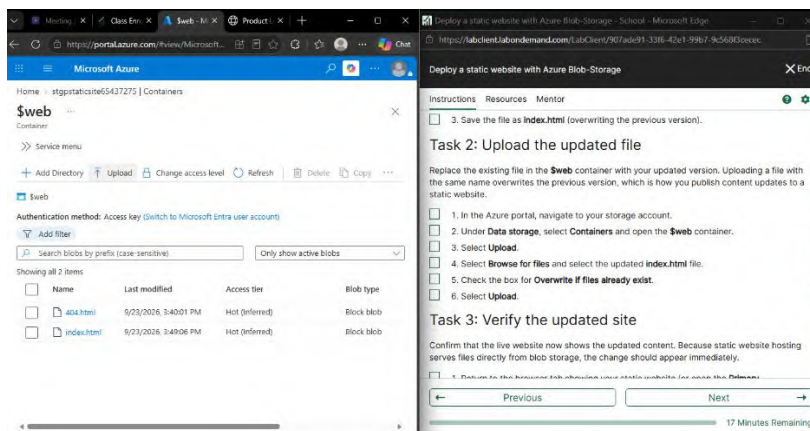
The user successfully verified their custom error page configuration by navigating to a non-existent URL on their static website and confirming that the intended "404 - Page Not Found" document is displayed.



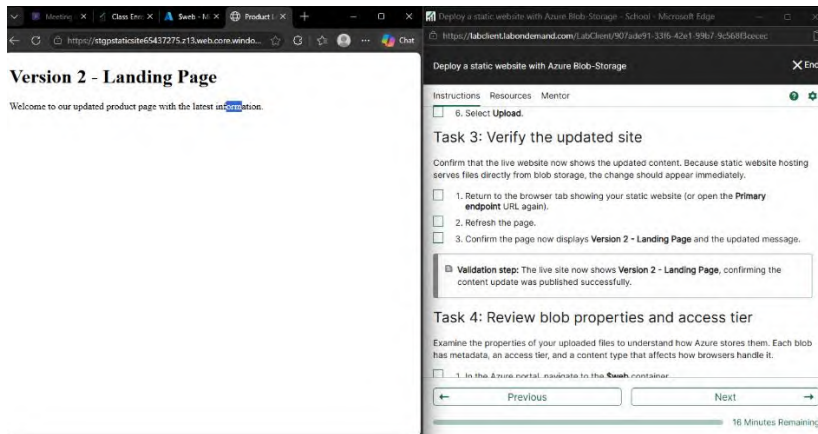
The user successfully navigated back from their custom error page to the main "Version 1 - Landing Page," completing the final verification step for their static website deployment lab.



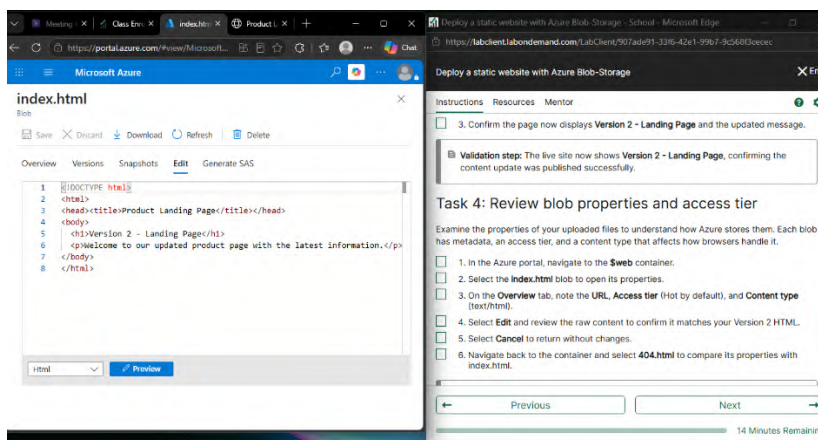
The user has updated the HTML code within the index.html file to display a new "Version 2 - Landing Page" heading and a revised welcome message.



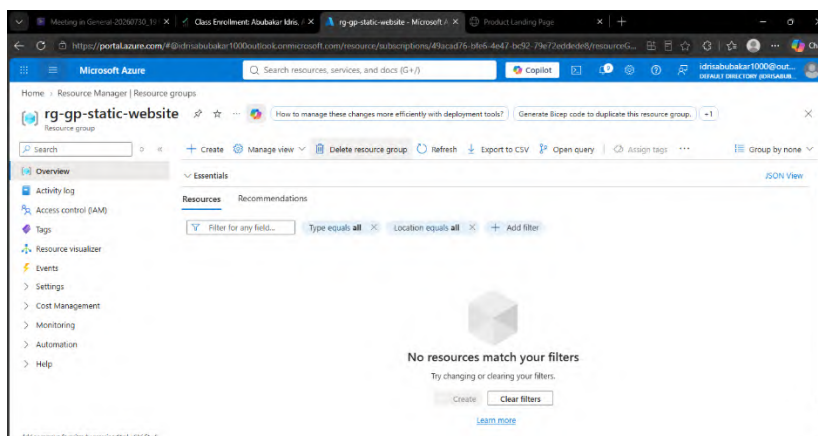
The user uploaded the modified index.html file to the Azure \$web storage container, overwriting the previous version to publish their content updates to the live static website.



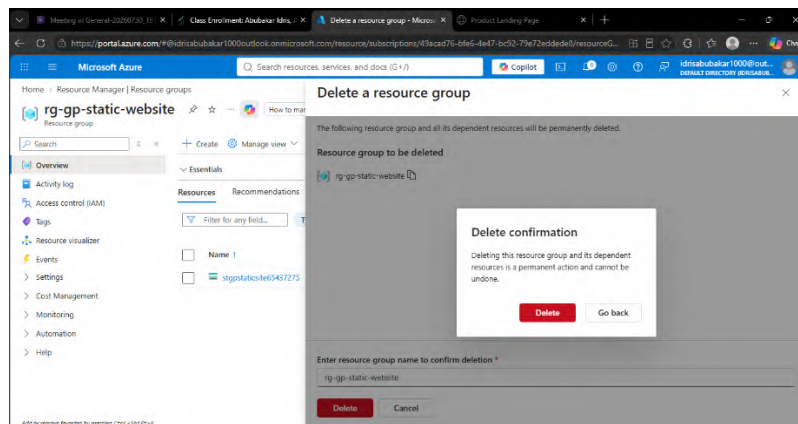
The user verified their content update by refreshing the live static website in the browser to confirm that the newly uploaded "Version 2" changes were successfully published.



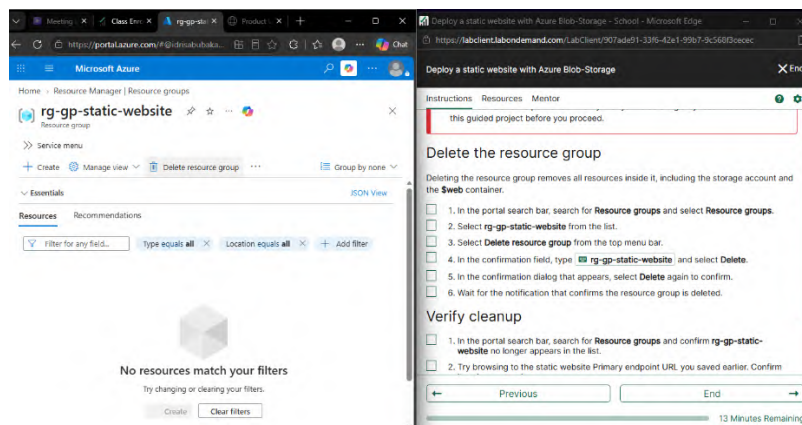
The user is reviewing the raw HTML content of the `index.html` blob within the Azure portal's edit interface to confirm it correctly matches their updated "Version 2" code, as directed by the lab instructions.



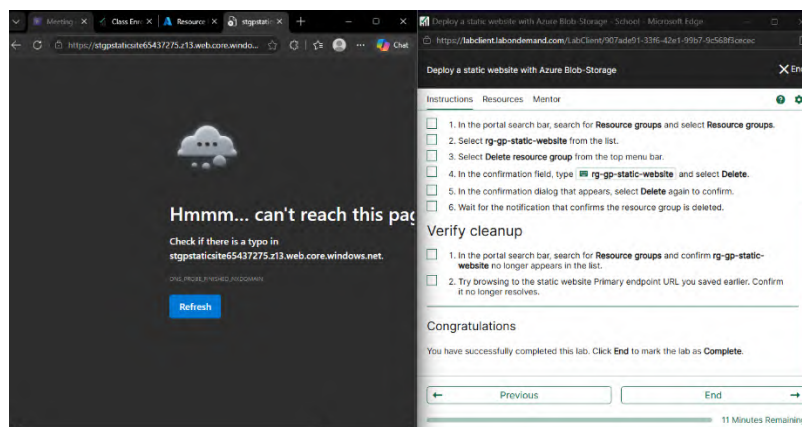
The user is viewing the rg-gp-static-website resource group in the Azure portal, which currently displays an empty state with no active resources present.



The user is confirming the permanent deletion of the rg-gp-static-website resource group and its dependent resources by entering its name into the final confirmation prompt within the Azure portal.

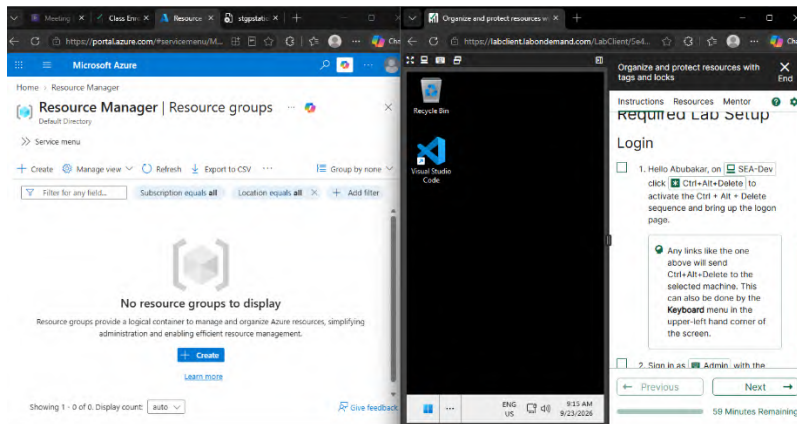


The user is executing the final lab cleanup phase by preparing to delete the rg-gp-static-website resource group and all its associated components to prevent unintended charges.

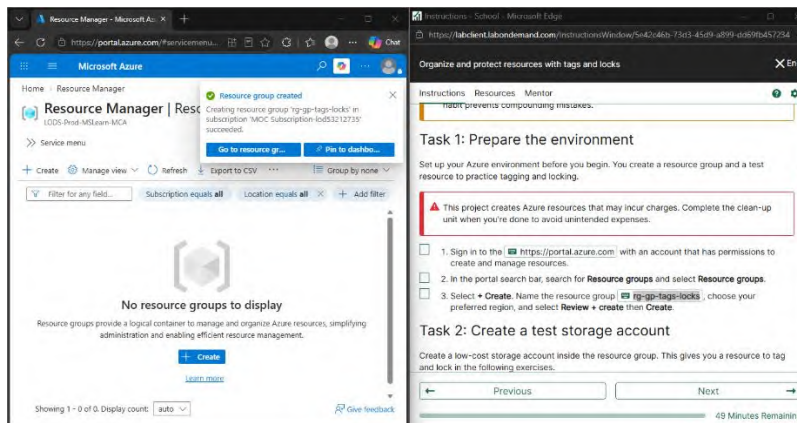


The user successfully verified the final lab cleanup step by attempting to browse to their static website's primary endpoint URL and confirming that the site no longer resolves, indicating all associated Azure resources were completely deleted.

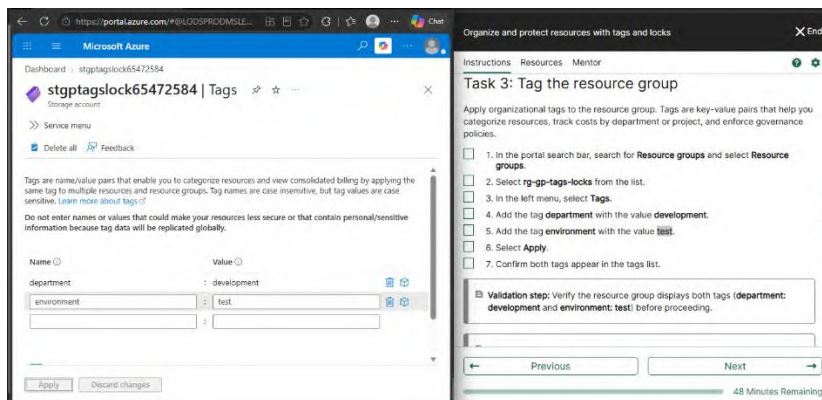
Organize and protect resources with tags and locks.



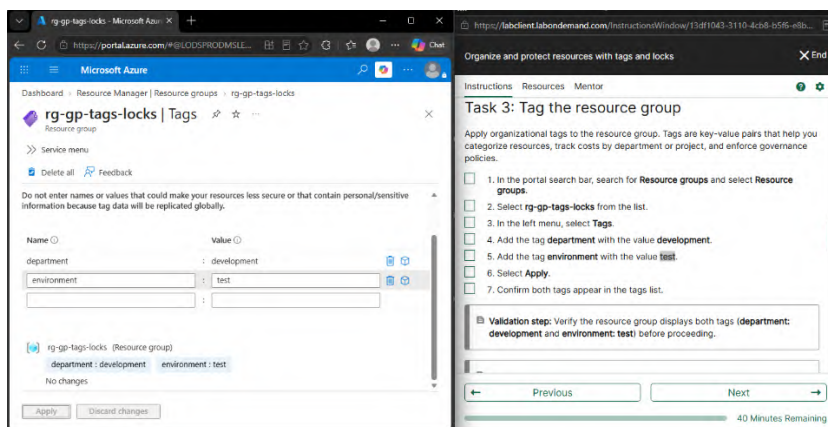
The user has initiated the setup for a new lab titled "Organize and protect resources with tags and locks" by accessing an empty Resource groups section in the Azure portal and successfully logging into the required Windows virtual machine environment.



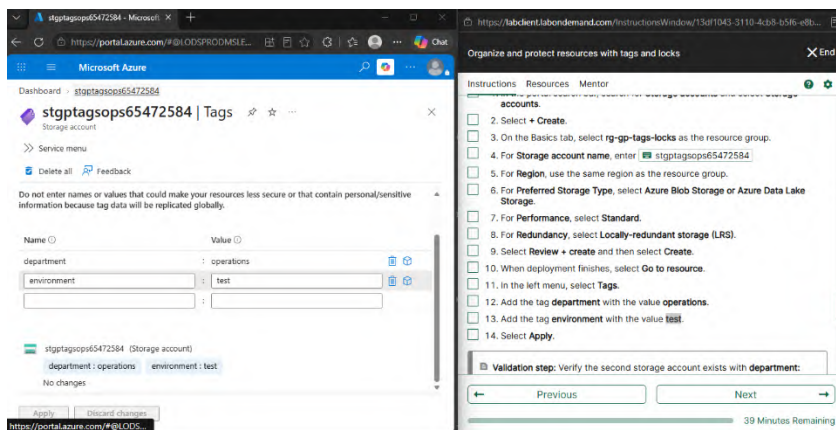
The user successfully completed Task 1 of the lab instructions by creating a new Azure resource group named rg-gp-tags-locks to prepare their environment for subsequent tagging and locking exercises.



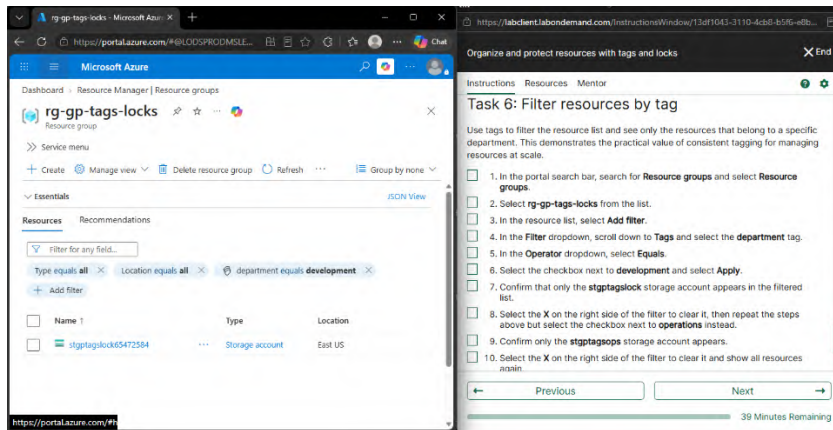
The user is following the lab instructions to apply organizational tags to their Azure resource by configuring the key-value pairs department: development and environment: test.



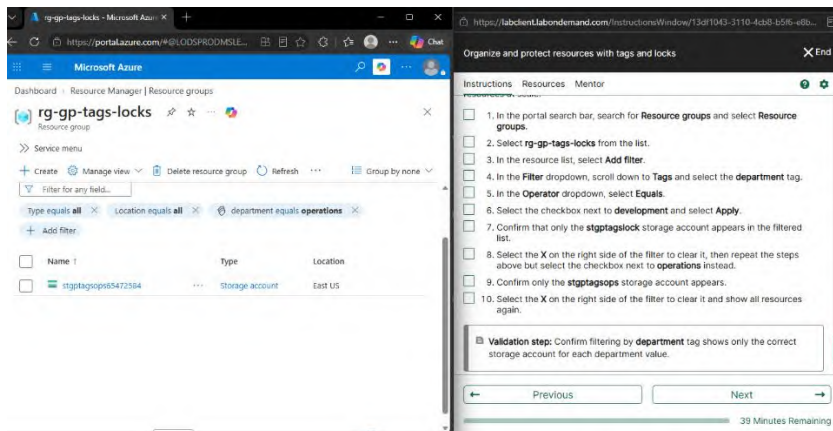
The user successfully completed the validation step for Task 3 by confirming that the department: development and environment: test tags were applied to the rg-gp-tags-locks resource group.



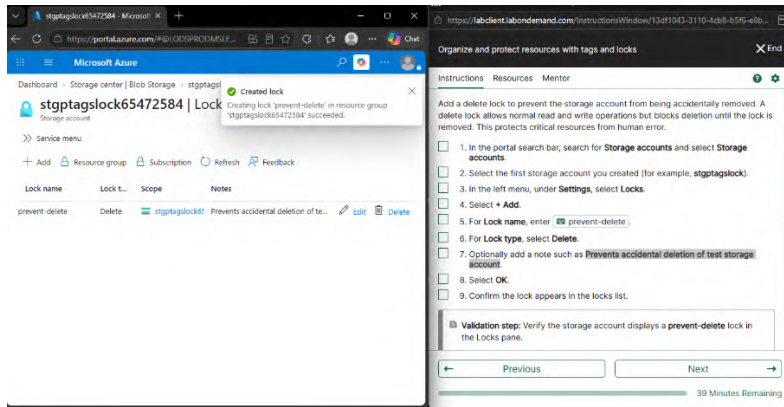
The user has successfully applied the organizational tags department: operations and environment: test to a newly created secondary storage account, completing the task's final configuration steps.



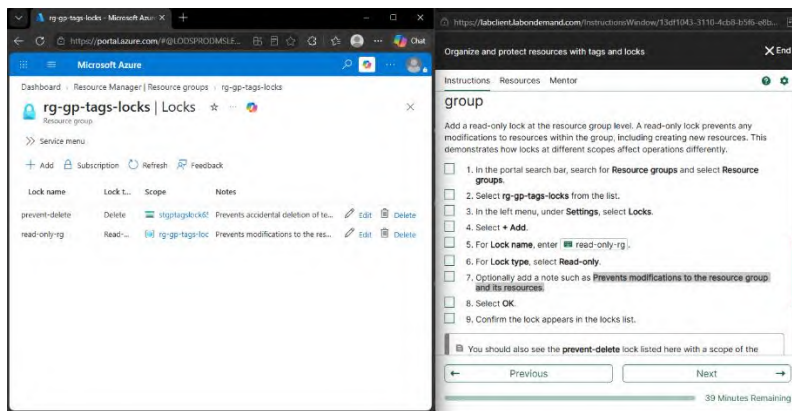
The user successfully filtered the resources within their resource group by applying the `department equals development` tag to isolate and view only the specific storage account associated with that department.



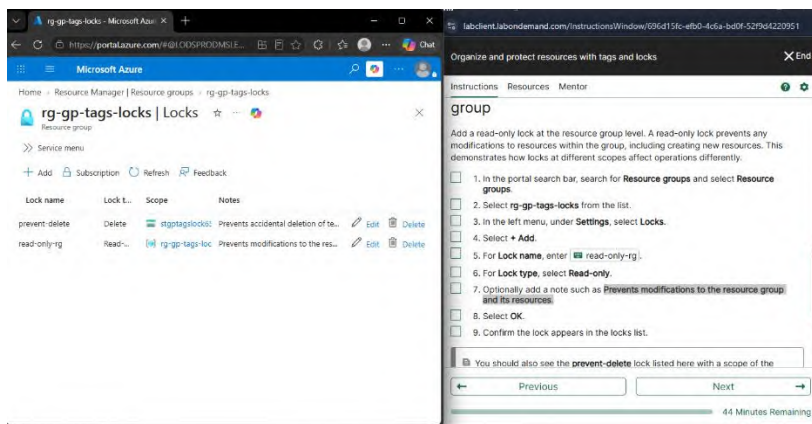
The user successfully applied a filter for the department equals operations tag to verify that only the corresponding secondary storage account is displayed in the resource list.



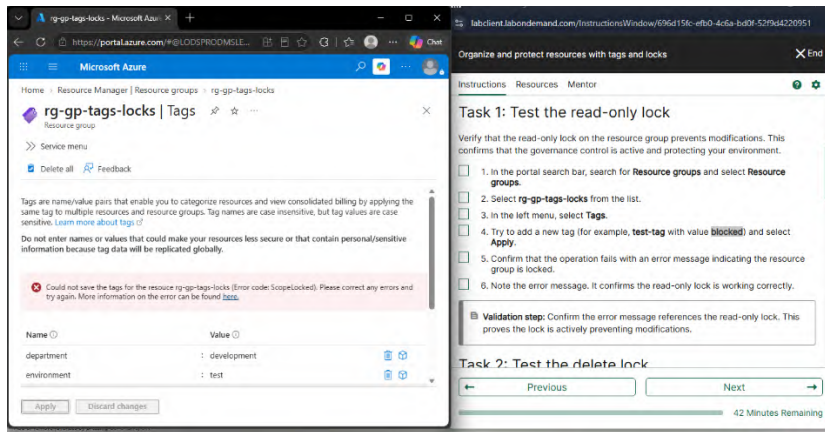
The user successfully applied a "Delete" lock named prevent-delete to their primary storage account to protect it from accidental removal, completing the validation step for the current lab task.



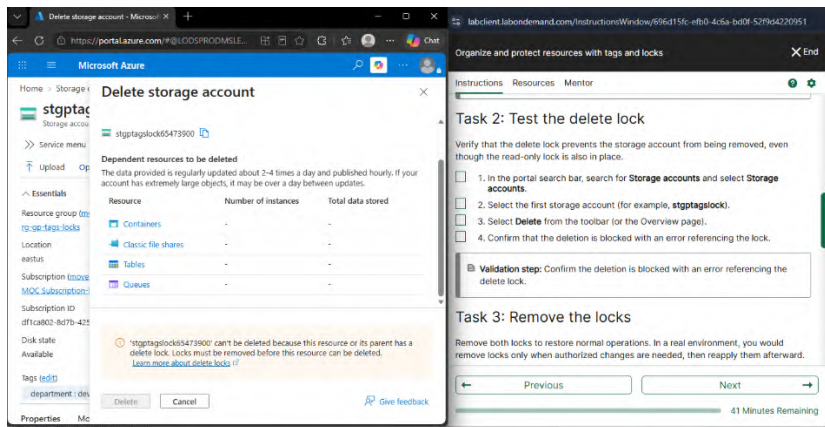
The user added a Read-only lock named read-only-rg at the resource group level to prevent modifications to all resources within rg-gp-tags-locks, confirming it alongside the existing prevent-delete lock in the Locks pane.



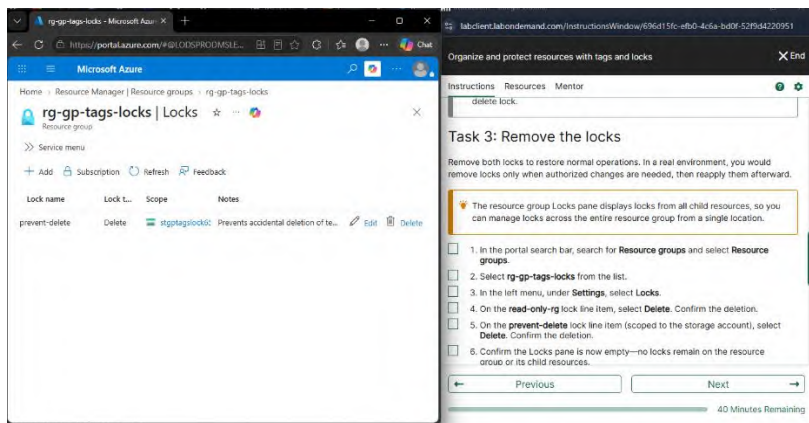
The user is completing the lab's validation step by checking the resource group's Locks pane to confirm that both the newly added read-only-rg lock and the previously configured prevent-delete lock are successfully listed together with their respective scopes.



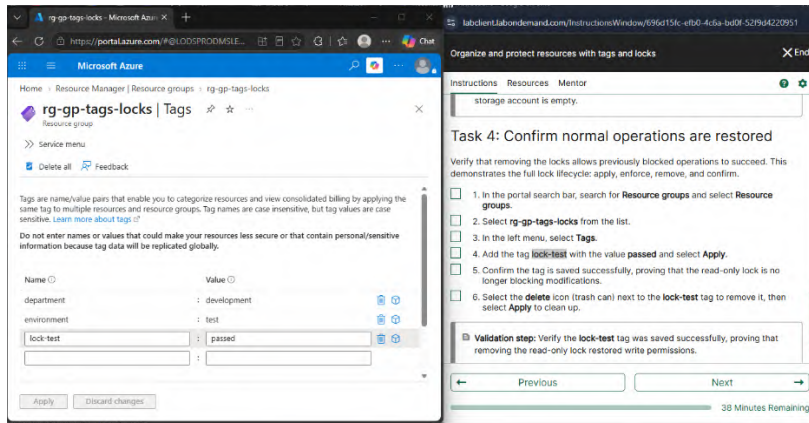
The user successfully tested the active read-only lock by attempting to add a new tag to the rg-gp-tags-locks resource group and confirming that the modification was blocked with a "ScopeLocked" error message.



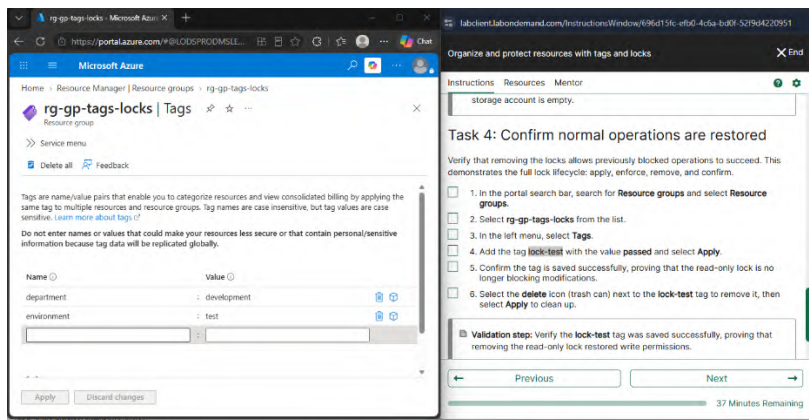
The user successfully tested the active delete lock by attempting to remove their primary storage account and confirming that the deletion was blocked by an error message referencing the lock.



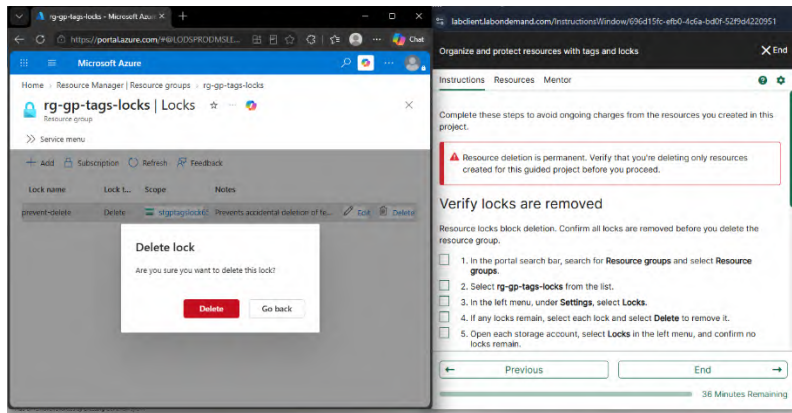
The user is following the lab instructions to remove the applied governance restrictions to restore normal operations, having successfully deleted the resource group's read-only lock and preparing to remove the remaining delete lock.



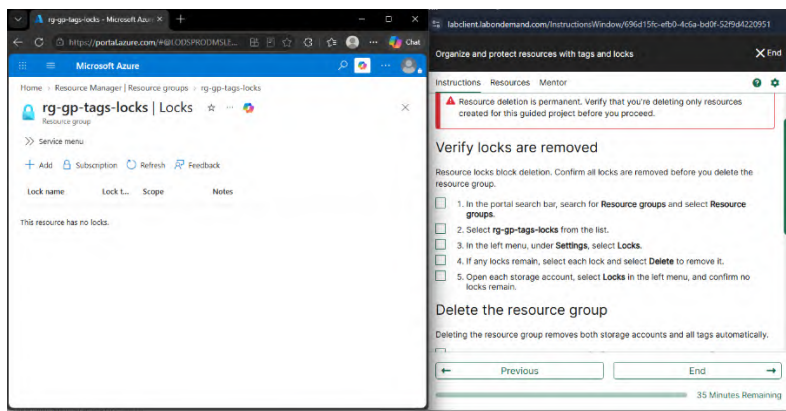
The user successfully added the lock-test: passed tag to the rg-gp-tags-locks resource group to verify that removing the read-only lock successfully restored normal write permissions.



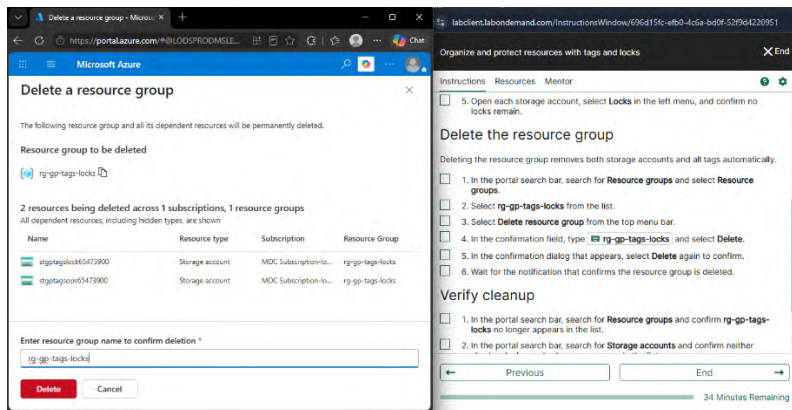
The user successfully executed the final cleanup step for Task 4 by deleting the temporary lock-test tag from the resource group to restore the original tag configuration.



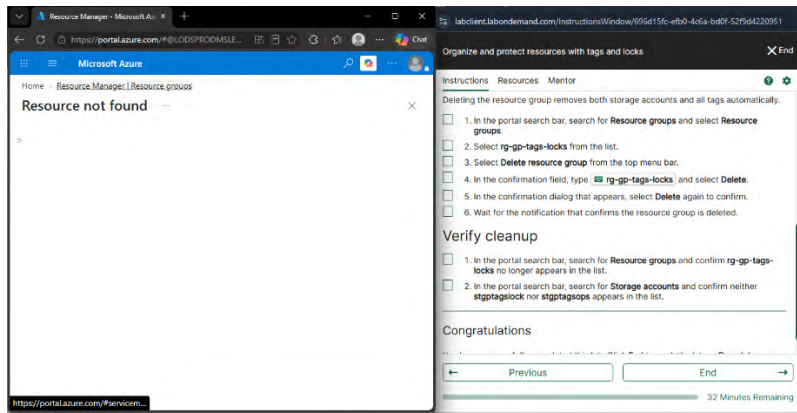
The user is confirming the removal of the remaining prevent-delete resource lock to ensure no locks block the final cleanup and deletion of the resource group.



The user has successfully verified that all resource locks have been removed from the rg-gp-tags-locks resource group by confirming that the Locks pane is empty, preparing the environment for final deletion.

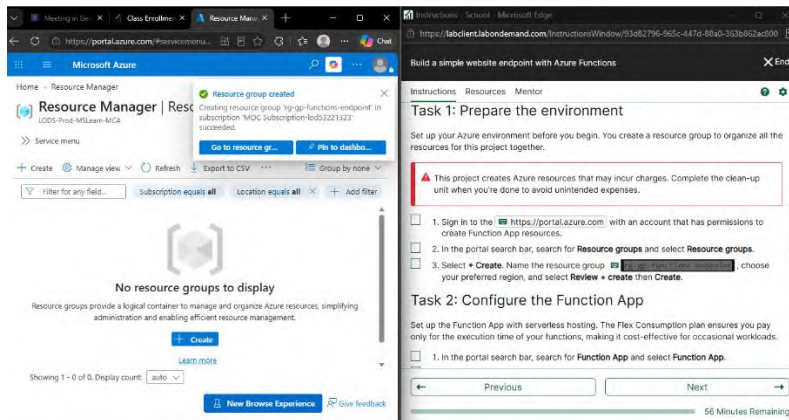


The user is executing the final lab cleanup step by typing the rg-gp-tags-locks resource group name to confirm the permanent deletion of the group and its dependent storage accounts.

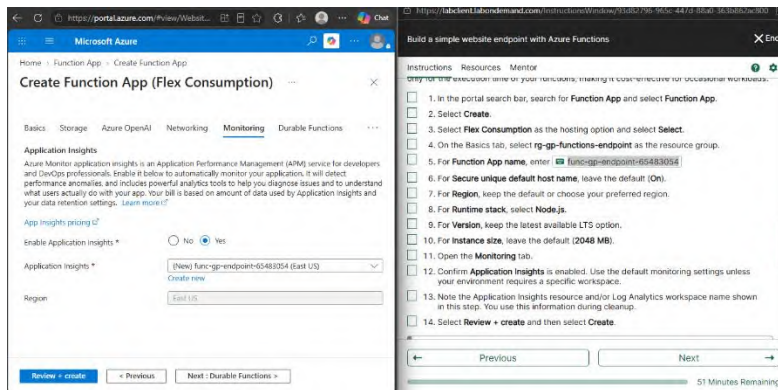


The user successfully completed the lab by verifying the final cleanup step, confirming that the resource group was completely deleted as evidenced by the "Resource not found" message in the Azure portal.

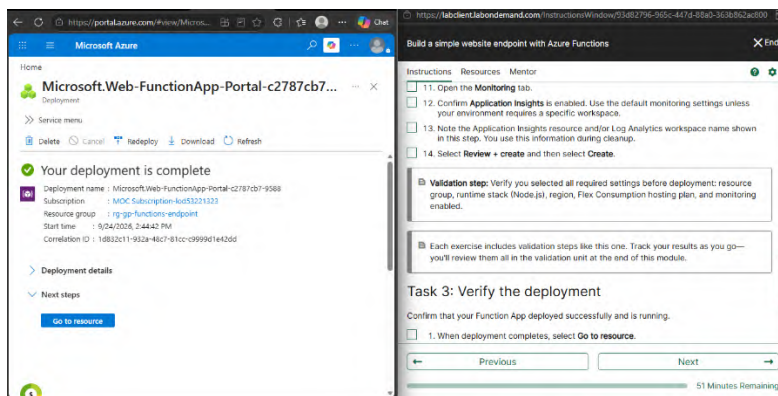
Build a simple website endpoint with Azure Functions.



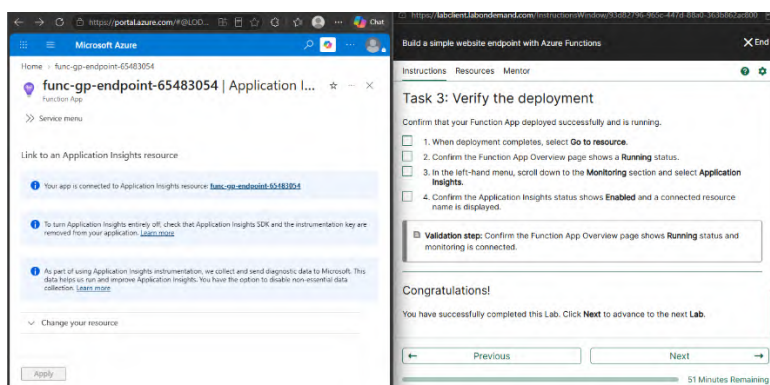
The user successfully completed the first task of a new lab by creating a resource group named rg-gp-functions-endpoint to prepare the environment for building an Azure Functions website endpoint.



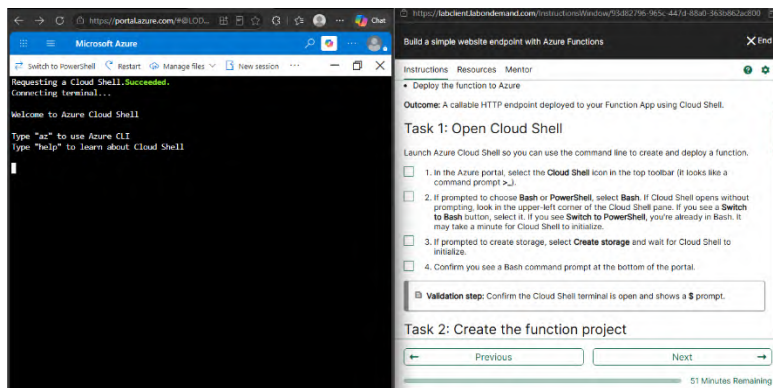
The user is configuring a new Azure Function App using a Flex Consumption plan and is currently on the Monitoring tab verifying that Application Insights is enabled before finalizing the creation of the resource.



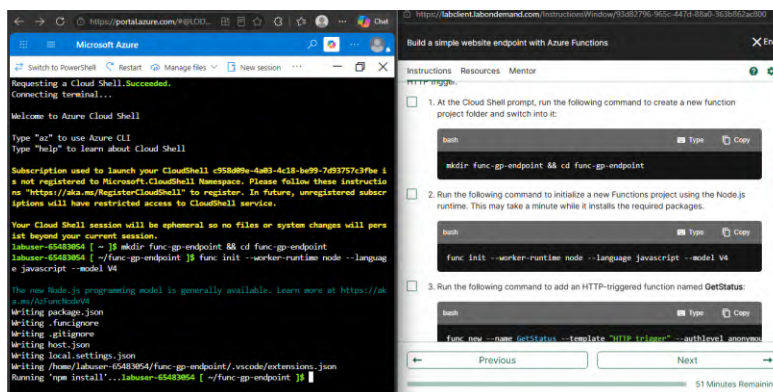
The user has successfully completed the deployment of their Azure Function App and is preparing to verify it by navigating to the newly created resource.



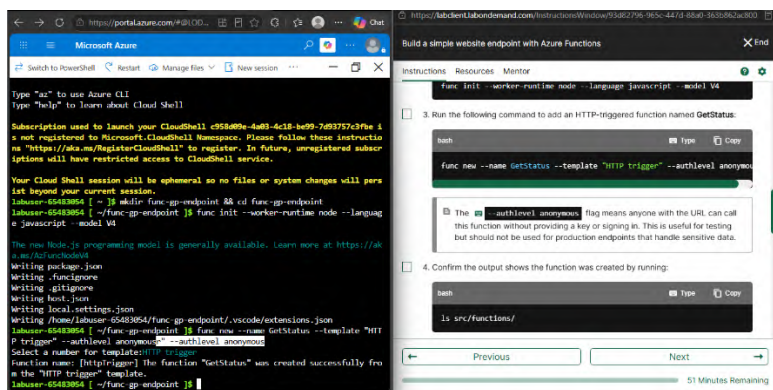
The user successfully completes the deployment verification task by confirming that their newly created Azure Function App is actively connected to the Application Insights monitoring resource.



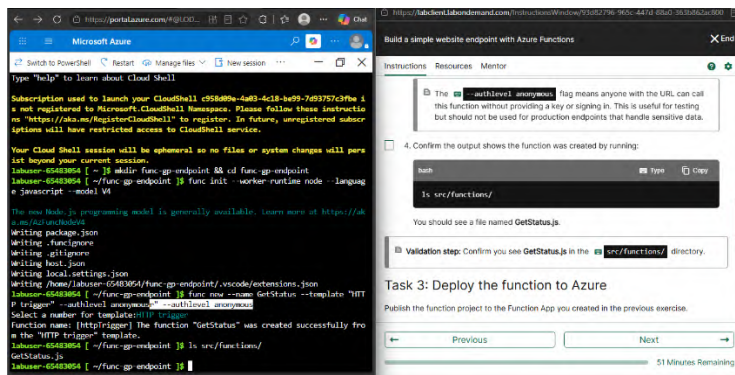
The user successfully launched and initialized the Azure Cloud Shell within the Azure portal to prepare for creating and deploying an Azure function via the command line.



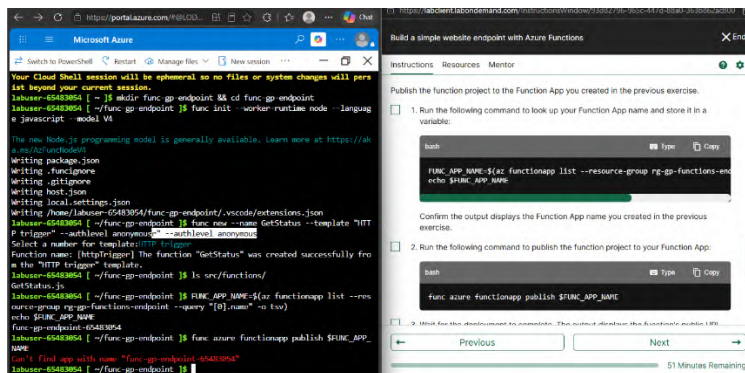
The user successfully used the Azure Cloud Shell to create a new project folder and initialize an Azure Functions project configured with the Node.js runtime.



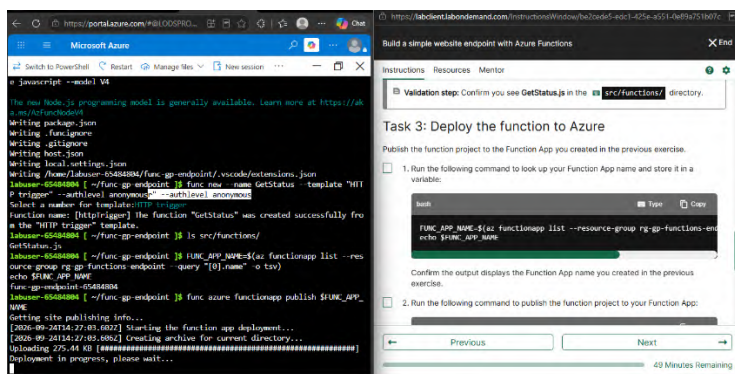
The user successfully executed a command in the Azure Cloud Shell to create a new HTTP-triggered function named `GetStatus` with anonymous access within their initialized project.



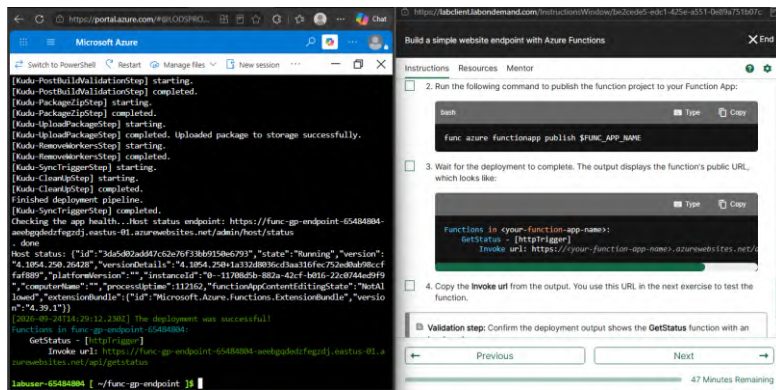
The user successfully executed the `ls src/functions/` command in the Azure Cloud Shell to confirm the creation of the `GetStatus.js` file, completing the validation step for the current lab task.



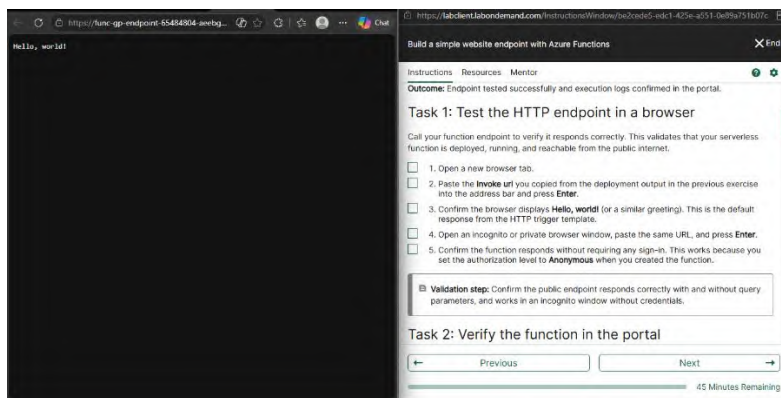
The user retrieved and stored their Function App name into a variable, but encountered a "Can't find app" error when attempting to execute the command to publish their local function project to Azure.



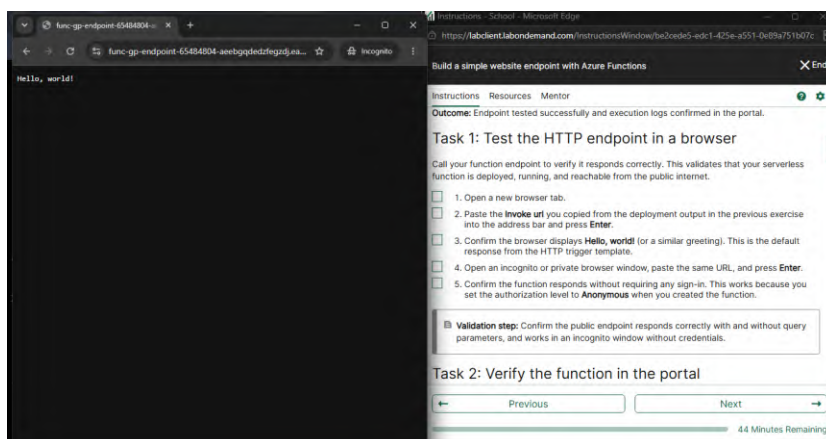
The user successfully executed the command to publish their local function project to their Azure Function App, with the terminal showing that the upload and deployment process is actively in progress.



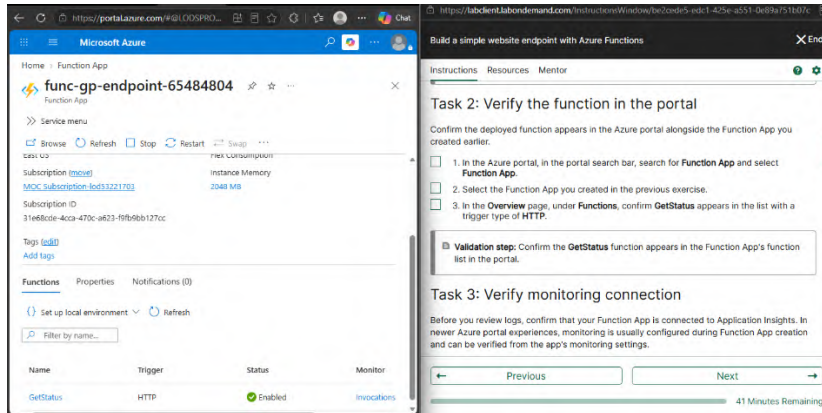
The user successfully completed the deployment of their function project to Azure, as evidenced by the successful deployment message in the terminal and the display of the public Invoke URL for the `GetStatus` function.



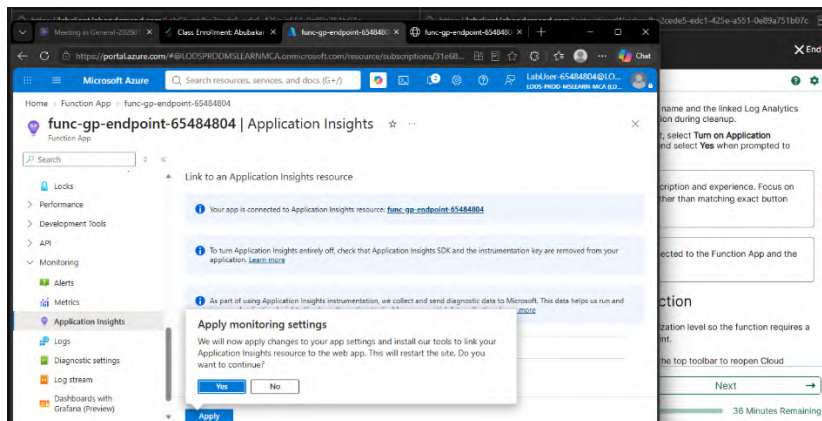
The user successfully tested their newly deployed serverless HTTP endpoint by navigating to its public Invoke URL in a web browser and confirming that it returned the expected "Hello, world!" greeting.



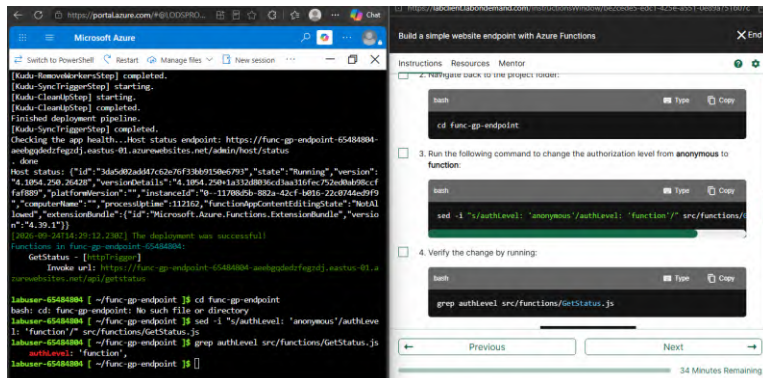
The user successfully tested the HTTP-triggered function in a private incognito browser window, confirming that the endpoint is publicly accessible without requiring credentials because it was created with an anonymous authorization level.



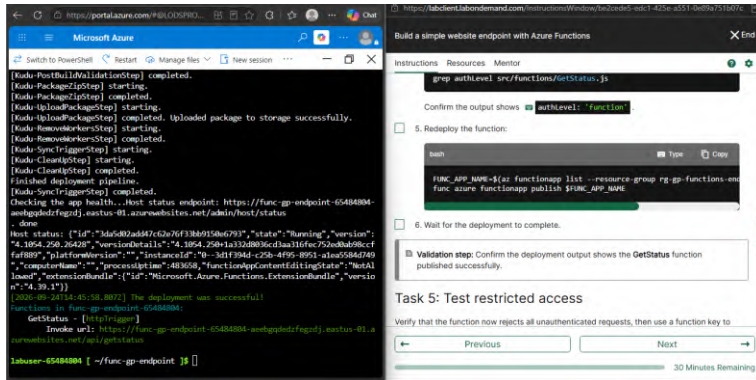
The user verified the deployment by checking the Azure portal and confirming that the 'GetStatus' function appears in the Function App's function list with an HTTP trigger type and an enabled status.



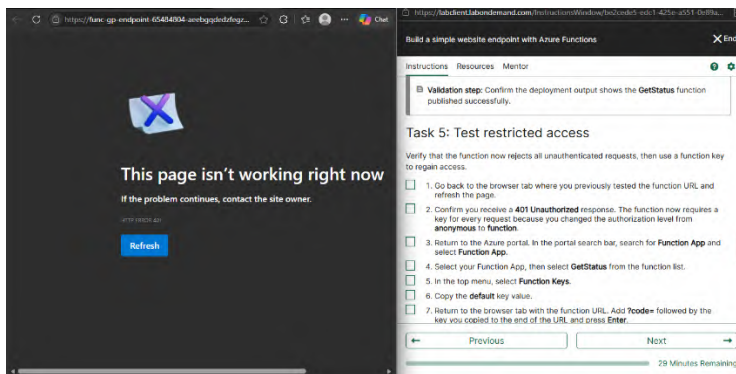
The user is applying monitoring settings in the Azure portal to link the Application Insights resource to their Function App, triggering a confirmation dialog to restart the site.



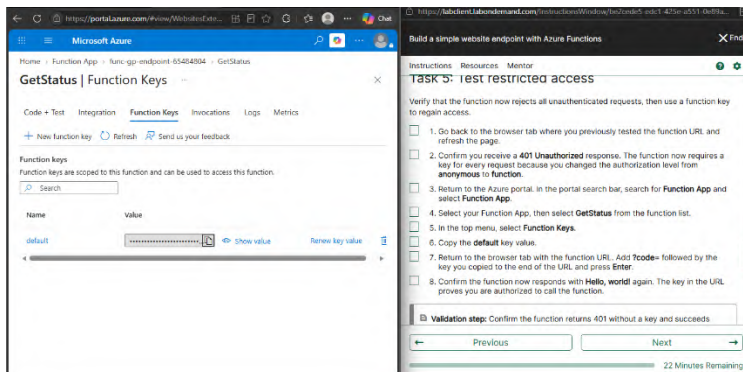
The user successfully executed a command in the Azure Cloud Shell to change the authorization level of their function from anonymous to function and verified the change using `grep`.



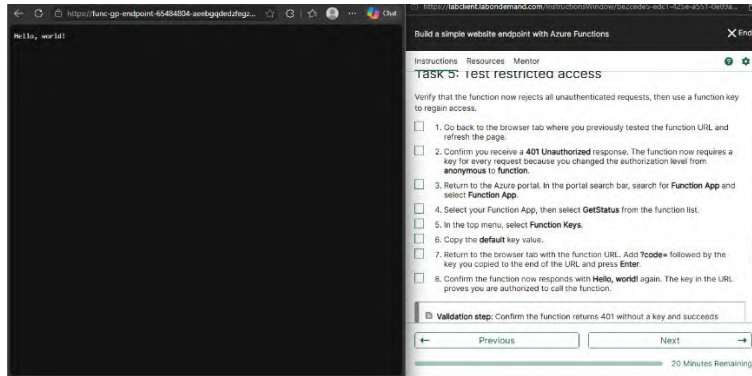
The user successfully redeployed the updated function project to Azure, confirming a successful deployment output and preparing to test restricted access for the function.



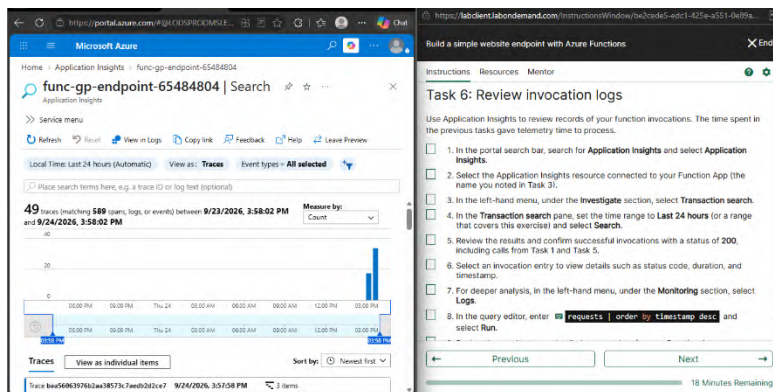
The user successfully tested restricted access by refreshing the browser tab for the function URL and confirming that it returned an HTTP 401 Unauthorized error because the authorization level was updated from anonymous to function.



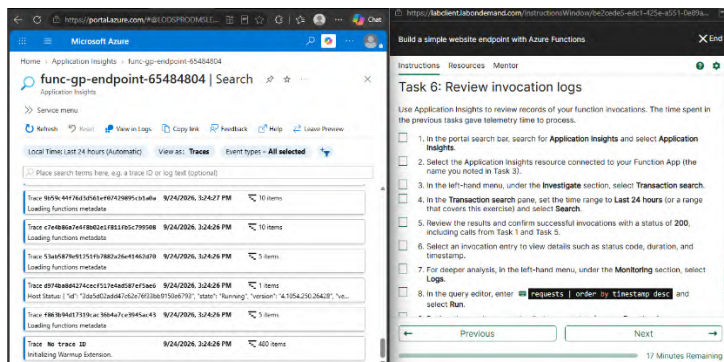
The user navigated to the Function Keys menu for the `GetStatus` function in the Azure portal to copy the `default` key value needed to authenticate and regain access to the endpoint.



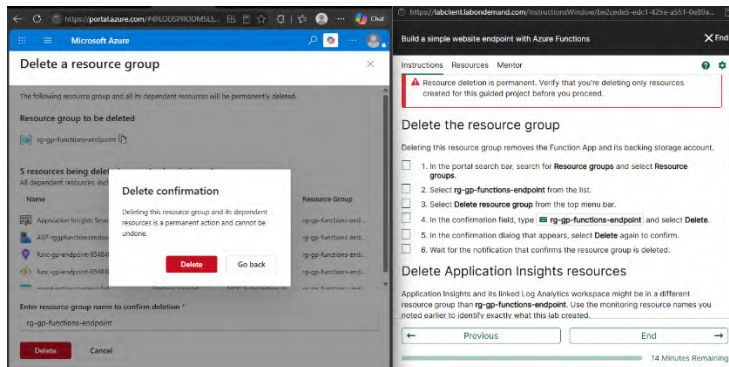
The user successfully tested the authenticated function endpoint by appending the copied default function key parameter to the URL, confirming that the browser successfully displays the "Hello, world!" greeting again.



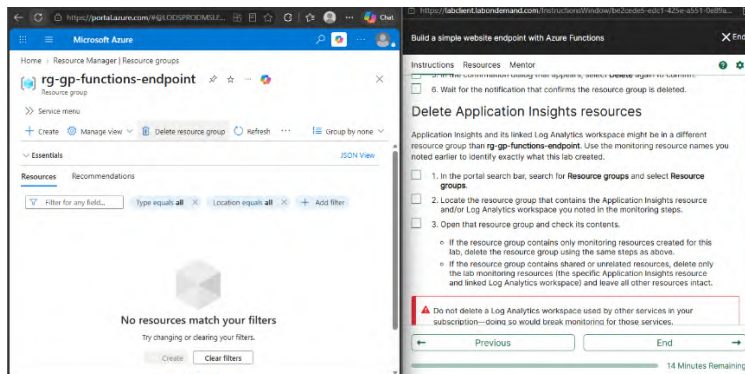
The user is reviewing the invocation logs and traces of their Azure Function within the Application Insights search pane to confirm successful function requests and responses.



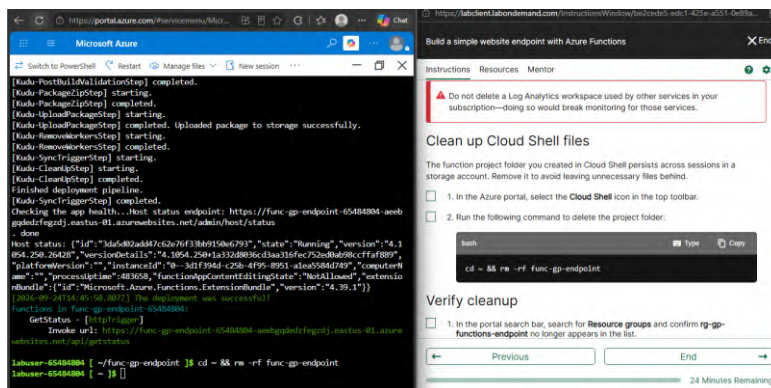
The user is examining detailed transaction traces and loading metadata logs within the Application Insights pane to inspect telemetry and execution records for their Azure Function.



The user is completing the lab cleanup by confirming the permanent deletion of the resource group and all its associated dependent resources in the Azure portal.

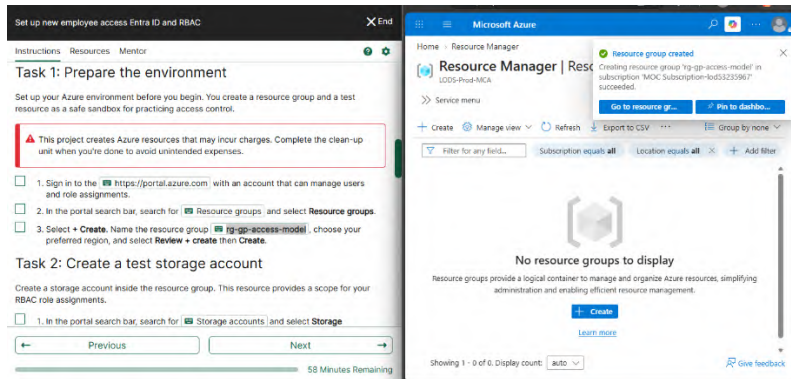


In the final step of the lab, the user successfully finished cleaning up their Azure environment by verifying that the resource group and all associated lab components were completely deleted.

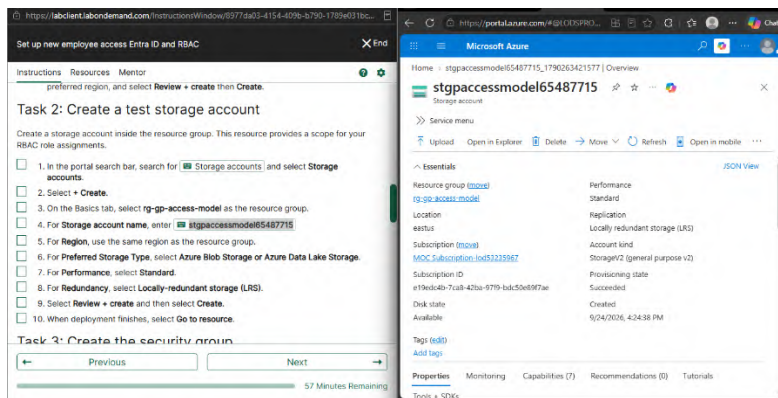


The user executed a command in the Azure Cloud Shell to clean up and remove the remaining project folder from storage, successfully finishing the final lab steps.

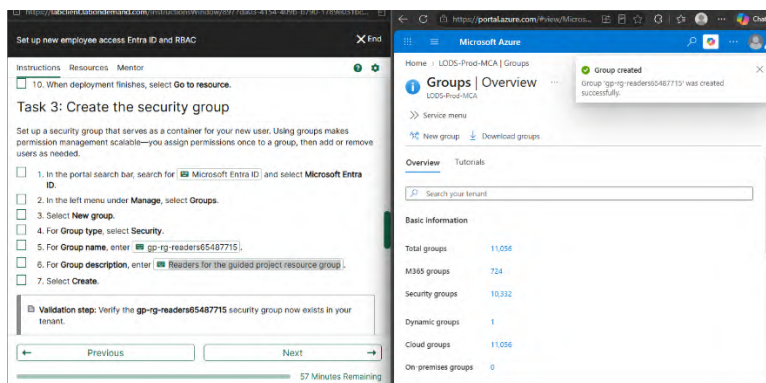
Set up new employee access Entra ID and RBAC.



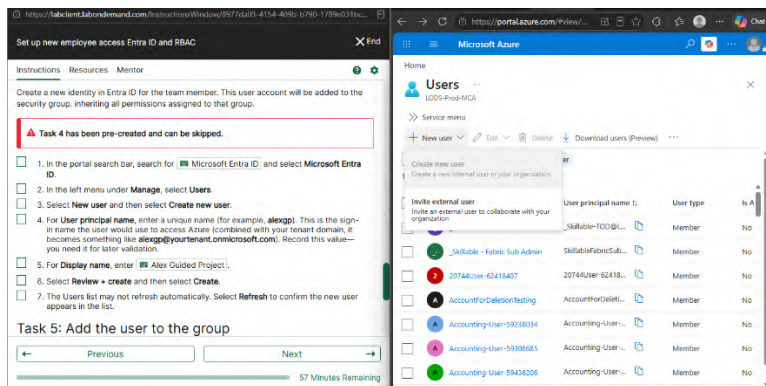
The user successfully created a new resource group named `rg-gp-access-model` in the Azure portal as part of preparing the environment for a new access control and RBAC lab.



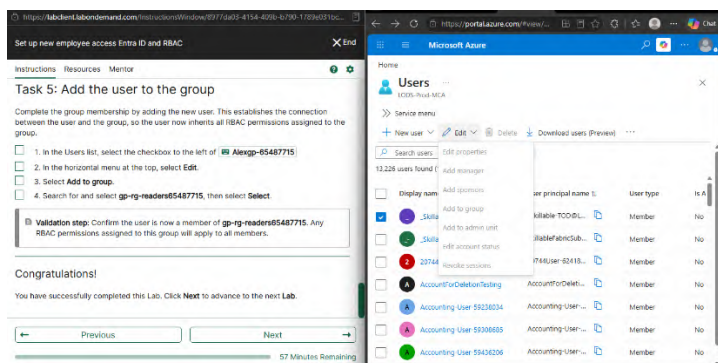
The user successfully created a new test storage account named `stgaccessmodel65487715` inside the `rg-gp-access-model` resource group in the Azure portal to serve as the scope for upcoming RBAC role assignments.



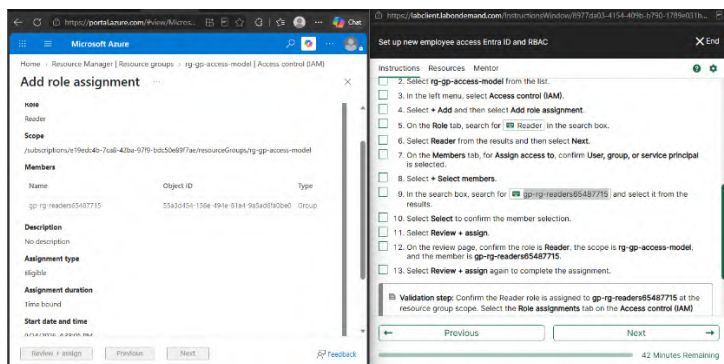
The user successfully created a new Microsoft Entra ID security group named `gp-rg-readers65487715` within their tenant to serve as a scalable container for managing user access permissions.



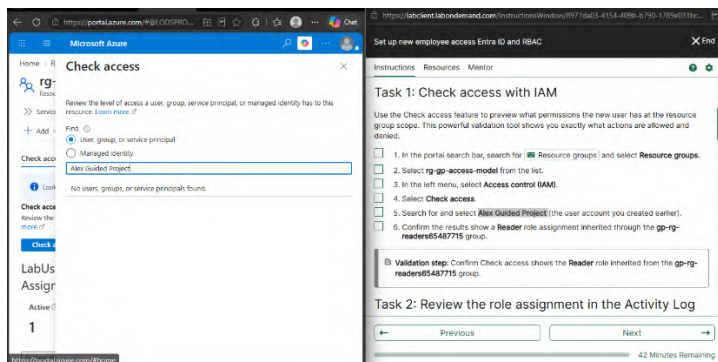
The user is navigating the Microsoft Entra ID portal to manage users and view the options for creating a new internal user or inviting an external team member.



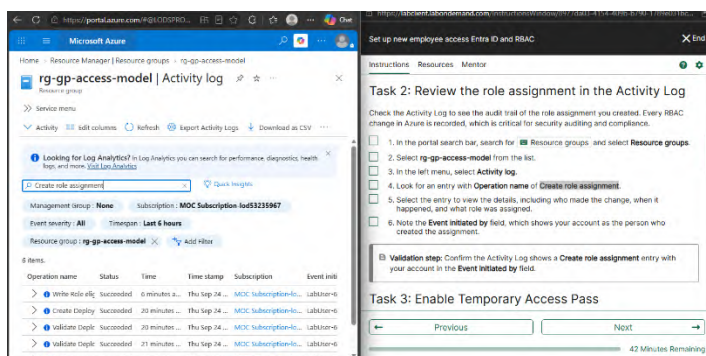
The user is adding the newly created user to the `gp-rg-readers65487715` security group via the Microsoft Entra ID portal to establish group membership and enable permission inheritance.



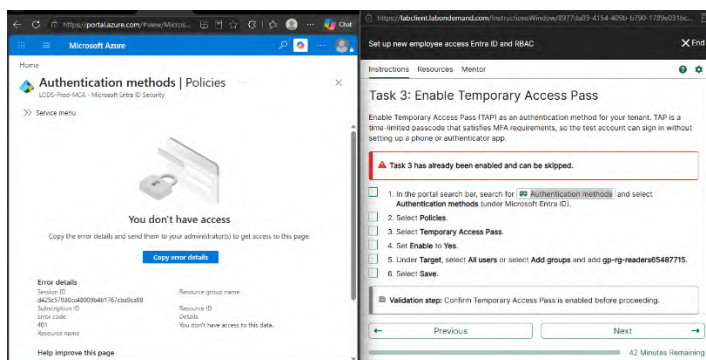
The user is configuring an RBAC role assignment in the Azure portal to assign the Reader role to the `gp-rg-readers65487715` security group at the `rg-gp-access-model` resource group scope.



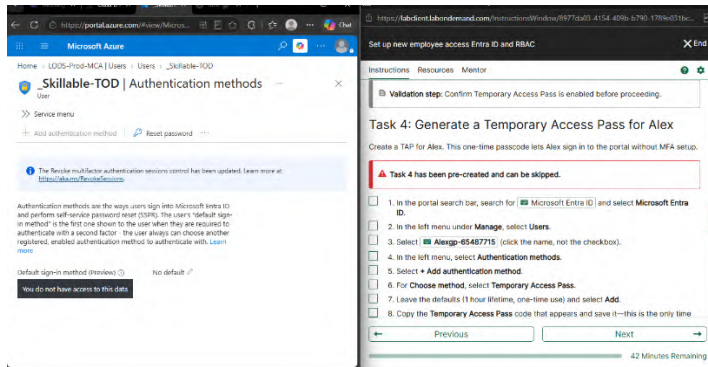
The user is using the "Check access" feature in the Azure portal to search for the newly created user and verify their inherited Reader role permissions through the security group.



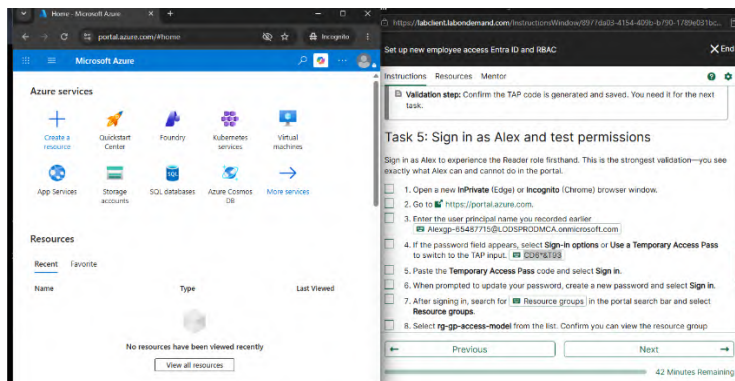
The user is reviewing the Activity Log in the Azure portal for the resource group to audit and verify the "Create role assignment" operation details and the account that initiated it.



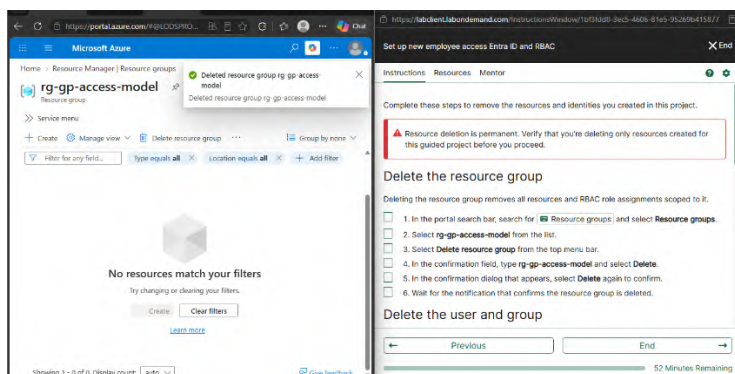
The user encountered an access denied error when attempting to configure the Temporary Access Pass authentication policies under Microsoft Entra ID.



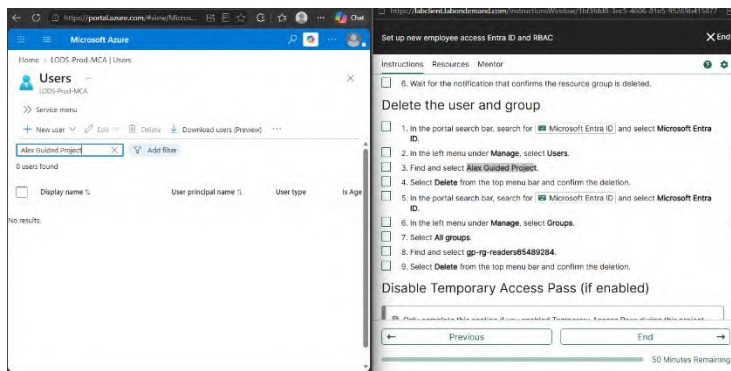
The user navigated to the user-specific authentication methods pane in the Microsoft Entra ID portal to proceed with generating a Temporary Access Pass, encountering an access restriction notice.



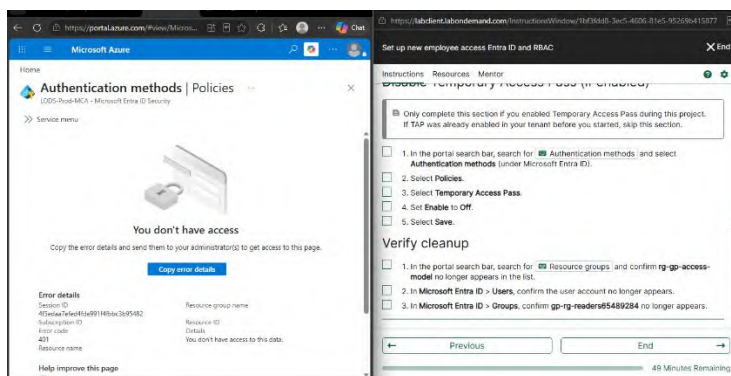
The user signed into an Incognito browser session as the new employee account using a Temporary Access Pass to test and verify the inherited Reader permissions within the Azure portal.



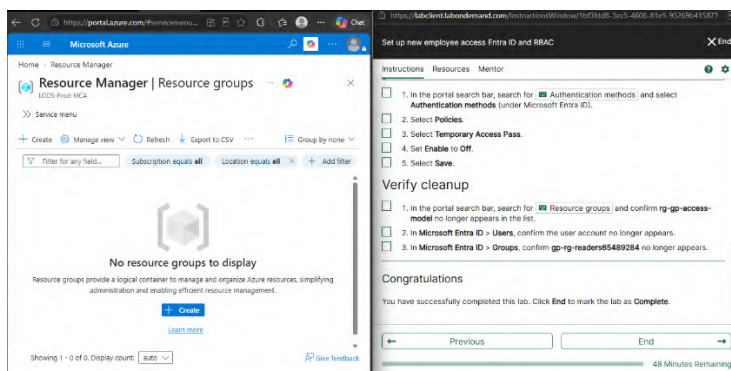
The user deleted the `rg-gp-access-model` resource group and its associated resources in the Azure portal as part of the final cleanup phase.



The user searched the Microsoft Entra ID users list to confirm the deletion of the test user account as part of the final cleanup steps.

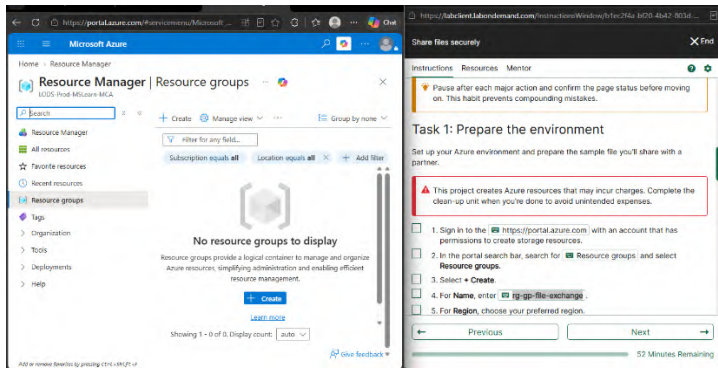


The user attempted to access the Temporary Access Pass policies under Microsoft Entra ID during the cleanup phase, encountering an access denied error message.

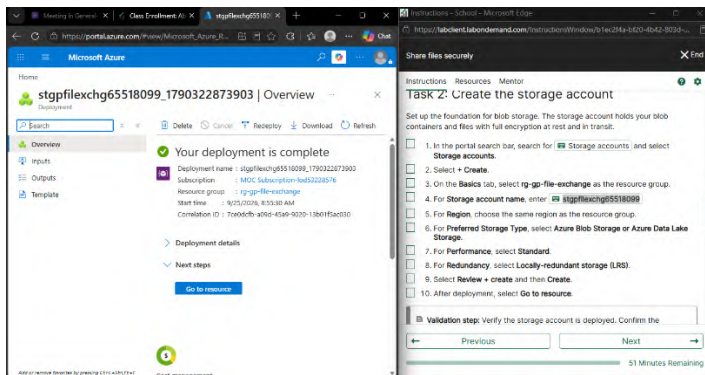


The user successfully verified the final cleanup stage in the Azure portal by confirming that the resource group no longer appears in the list, completing the access control and RBAC management lab.

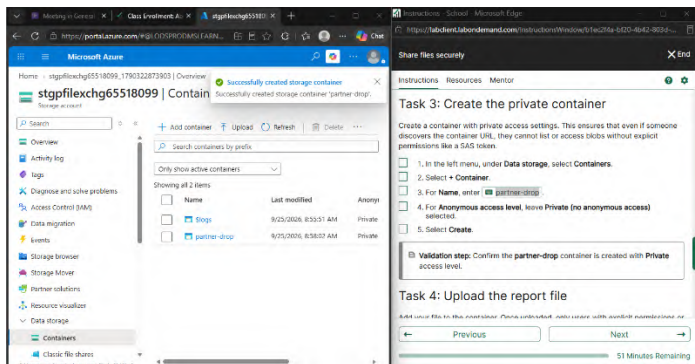
Share files securely.



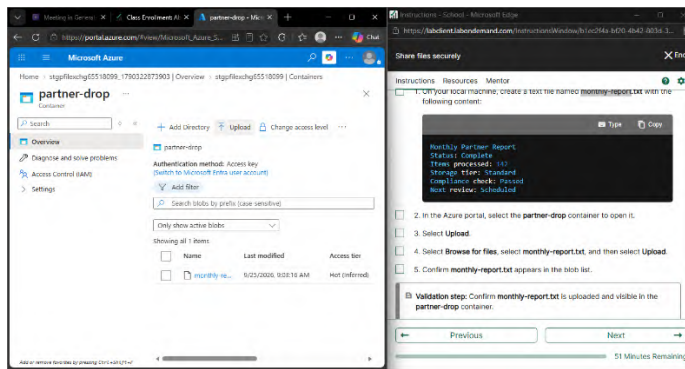
The user is preparing the environment for a new file-sharing project by navigating to the "Resource groups" pane in the Azure portal to create a dedicated resource group named `rg-gp-file-exchange`.



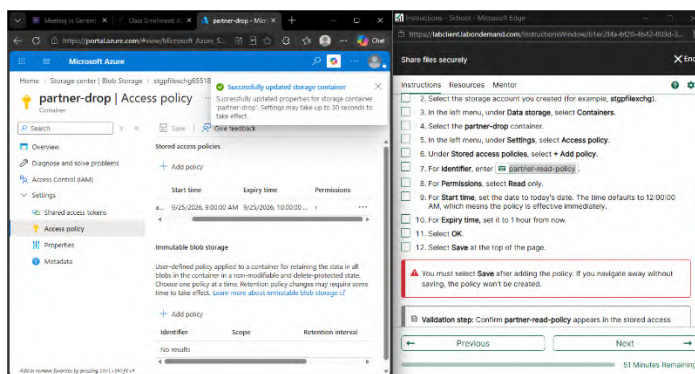
The user successfully completed the deployment of a new storage account named `stgpfilerch65518099` within the `rg-gp-file-exchange` resource group in the Azure portal.



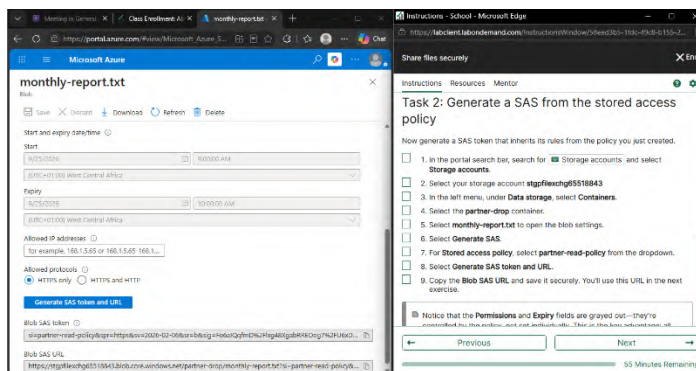
The user successfully created a new private storage container named `partner-drop` within the storage account to securely manage file sharing.



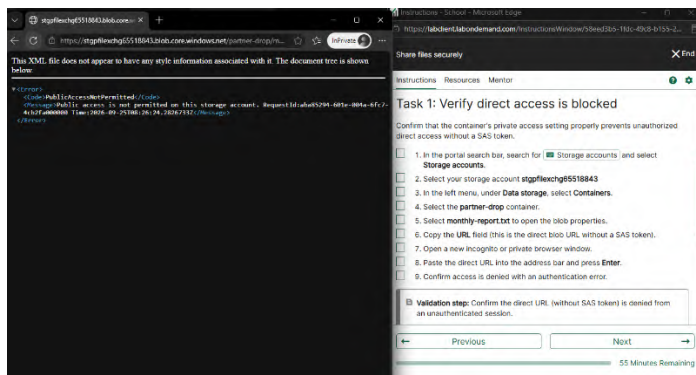
The user uploaded the `monthly-report.txt` file into the `partner-drop` storage container within the Azure portal and verified its appearance in the blob list.



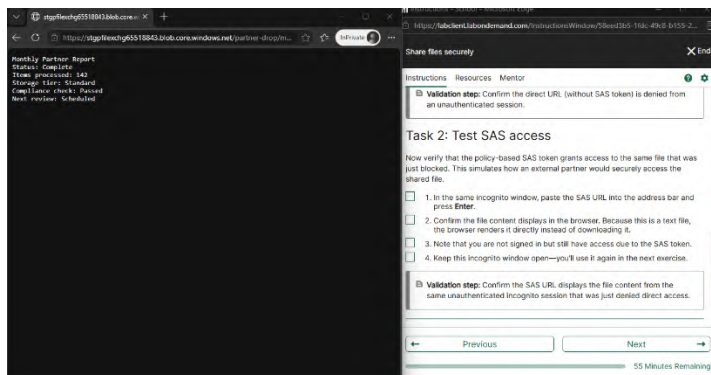
The user successfully created and saved a stored access policy named `partner-read-policy` with read-only permissions for the `partner-drop` storage container in the Azure portal.



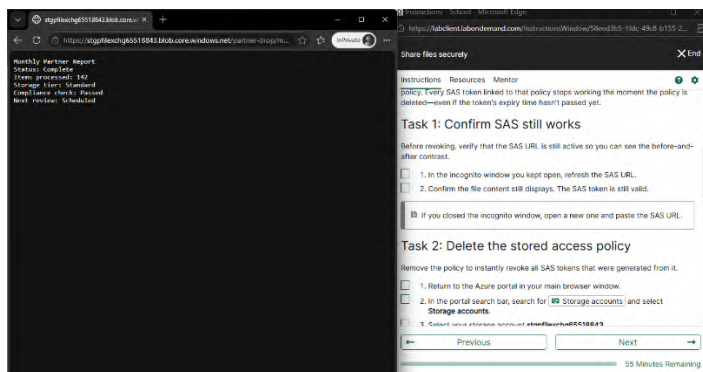
The user is generating a Blob SAS token and URL for the `monthly-report.txt` file by associating it with the previously created `partner-read-policy` stored access policy in the Azure portal.



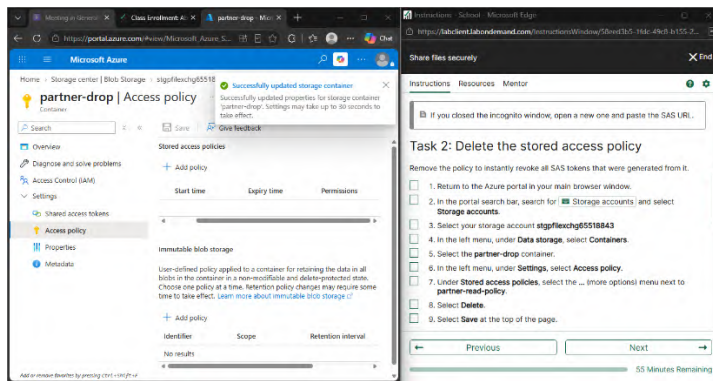
The user is testing the blob's direct URL without a SAS token in an InPrivate browser window, successfully confirming that public access is blocked and denied with an authentication error.



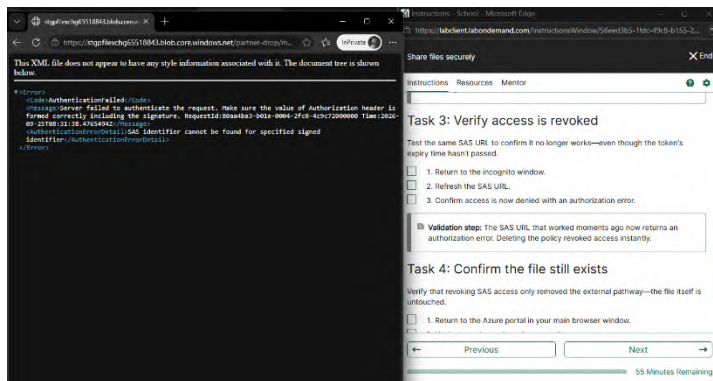
The user is testing the policy-based SAS URL in an unauthenticated InPrivate browser window, successfully verifying that the file contents of the shared report are displayed as expected.



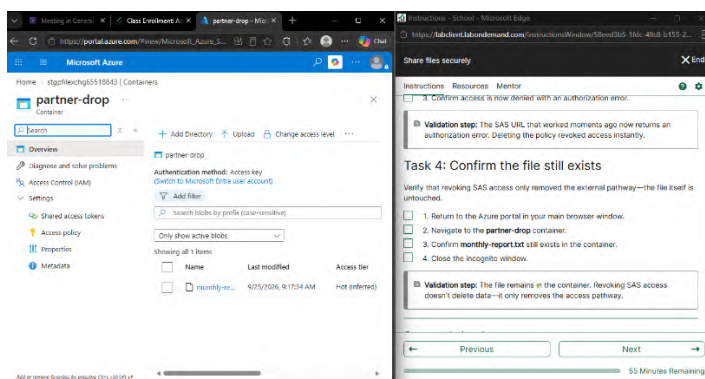
The user is refreshing the SAS URL in an InPrivate browser window to confirm that the shared file content is still accessible prior to revoking the access policy.



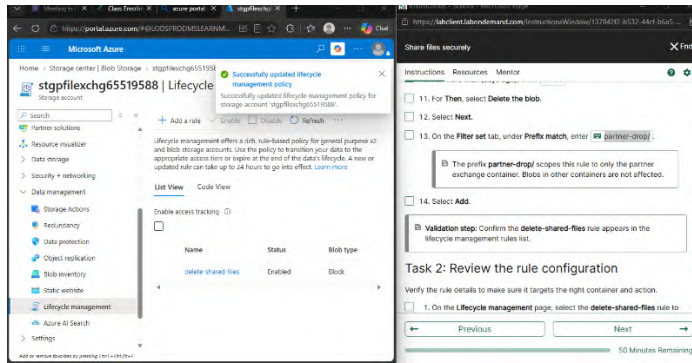
The user deleted the `partner-read-policy` stored access policy from the container settings in the Azure portal and saved the changes to instantly revoke all associated SAS tokens.



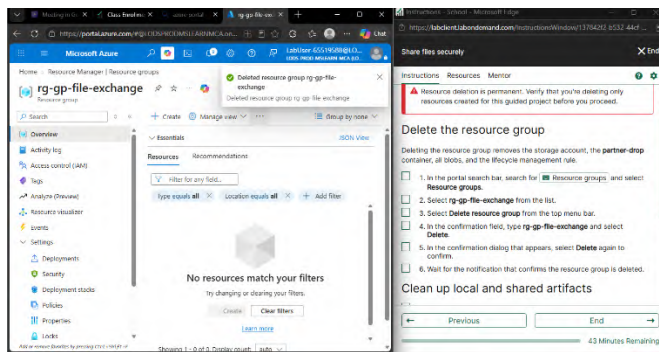
The user refreshed the SAS URL in the InPrivate browser window and successfully verified that access is now denied with an authentication error because the underlying access policy was deleted.



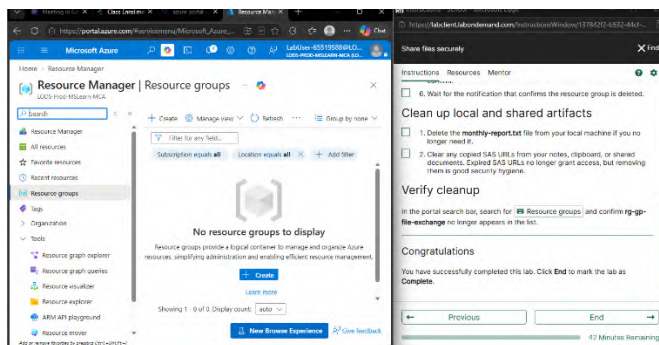
The user returned to the Azure portal and confirmed that `monthly-report.txt` still exists within the `partner-drop` container, demonstrating that revoking SAS access only removes the external access pathway rather than deleting the underlying file.



The user successfully created and enabled a lifecycle management policy named `delete-shared-files` within the storage account to automatically manage data retention and blob deletion for the `partner-drop` container.

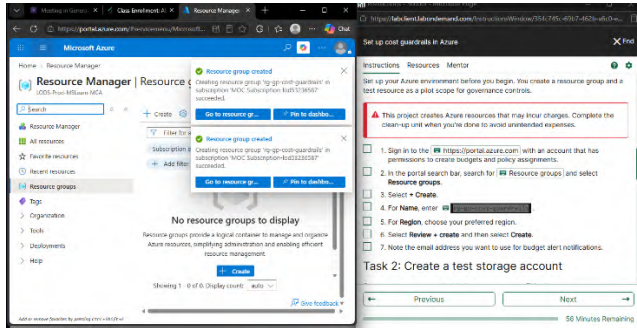


The user successfully deleted the `rg-gp-file-exchange` resource group and all associated storage resources in the Azure portal as part of the final project clean-up phase.

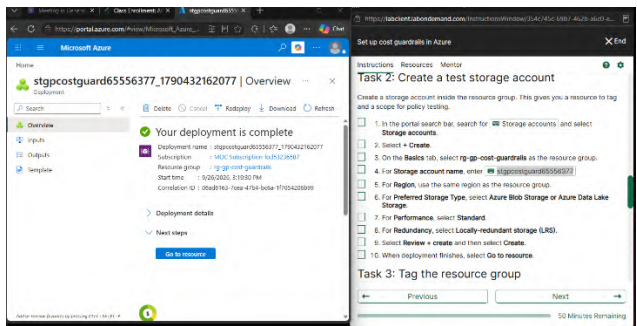


The user successfully completed the final verification step of the cleanup phase in the Azure portal by confirming that the `rg-gp-file-exchange` resource group no longer appears, marking the successful completion of the file-sharing lab.

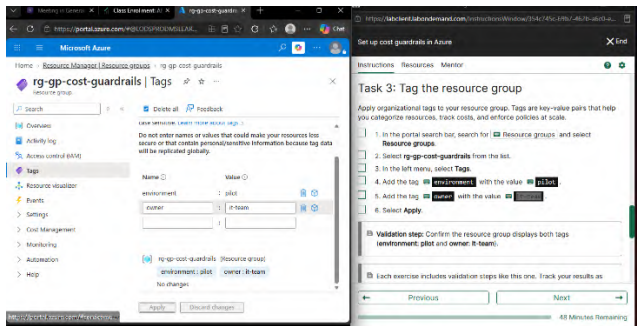
Set up cost guardrails in Azure.



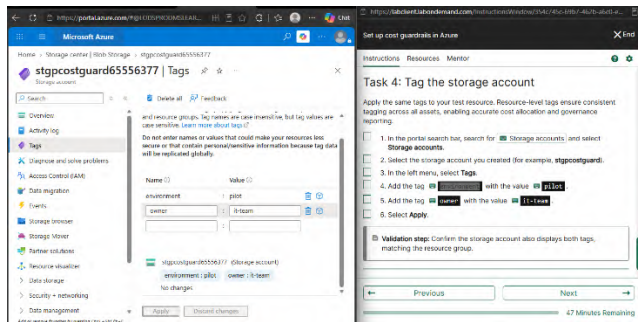
In the provided image, the user has successfully created a new resource group named `rg-gp-cost-guardrails` in the Azure portal as the initial step for setting up cost guardrails and governance controls.



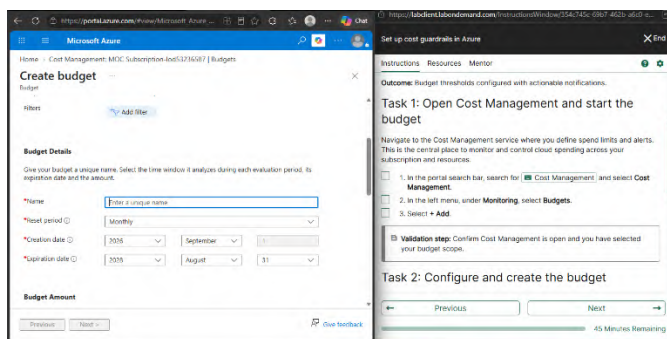
In the provided image, the user has successfully completed the deployment of a new test storage account named stgpcostguard65556377 within the rg-gp-cost-guardrails resource group to establish a testing scope for policy and governance controls.



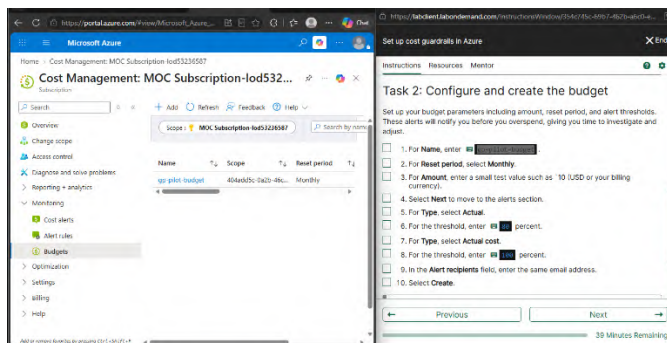
In the provided image, the user is applying organizational tags (environment: pilot and owner: it-team) to the rg-gp-cost-guardrails resource group within the Azure portal to support cost tracking and policy enforcement.



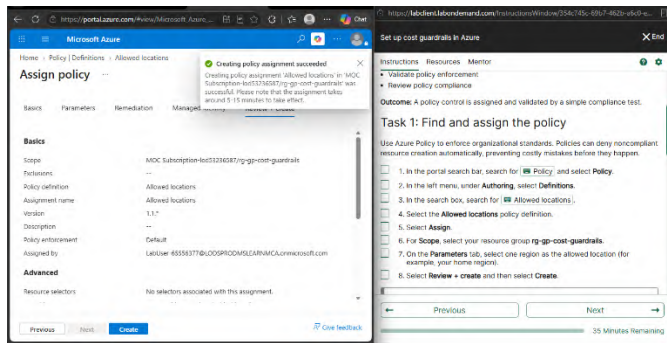
In the provided image, the user is applying the same organizational tags (environment: pilot and owner: it-team) to the test storage account within the Azure portal to ensure consistent cost allocation and governance.



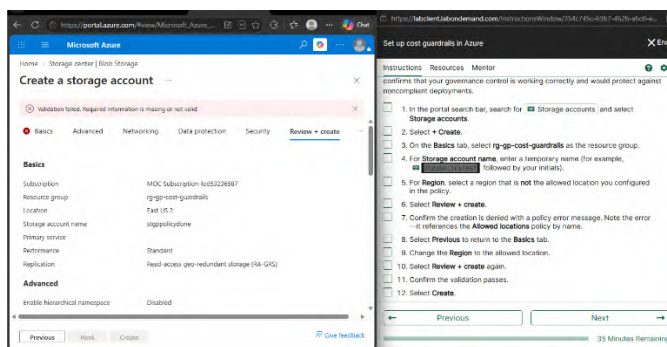
In the provided image, the user is navigating to the Cost Management Budgets section in the Azure portal and initiating the creation of a new budget by configuring its basic details and timeframe.



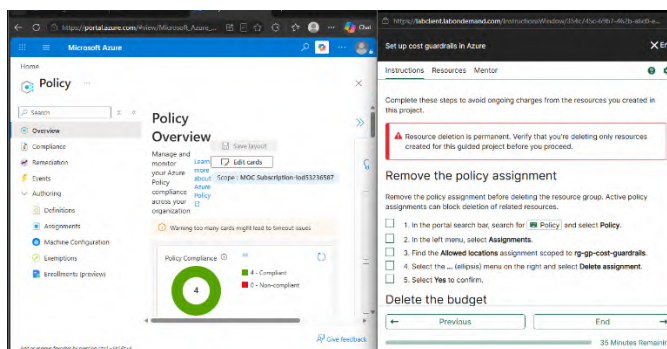
In the provided image, the user has successfully created and listed a new monthly budget named `gp-pilot-budget` within the Cost Management dashboard in the Azure portal.



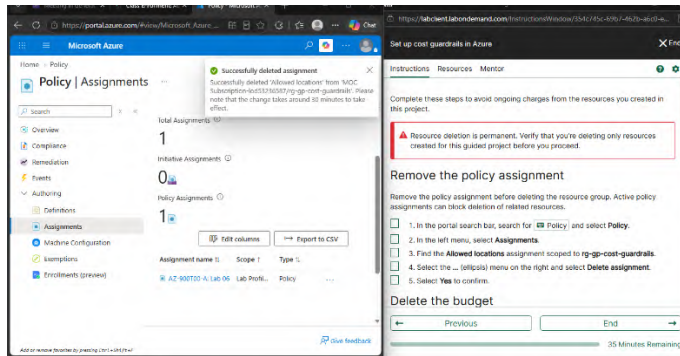
In the provided image, the user successfully created an "Allowed locations" policy assignment scoped to the rg-gp-cost-guardrails resource group in the Azure portal.



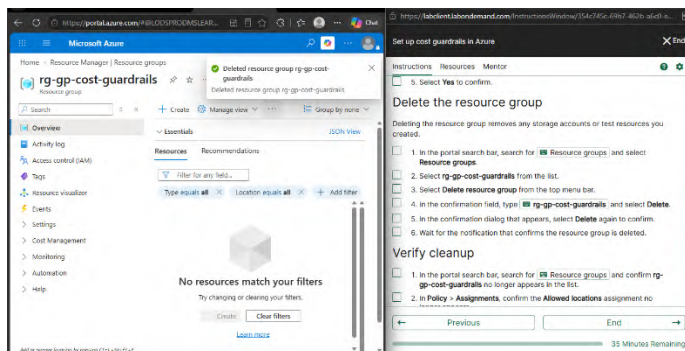
In the provided image, the user attempted to create a test storage account in a non-compliant region within the Azure portal, resulting in a validation failure that successfully verified the enforcement of the Allowed locations policy.



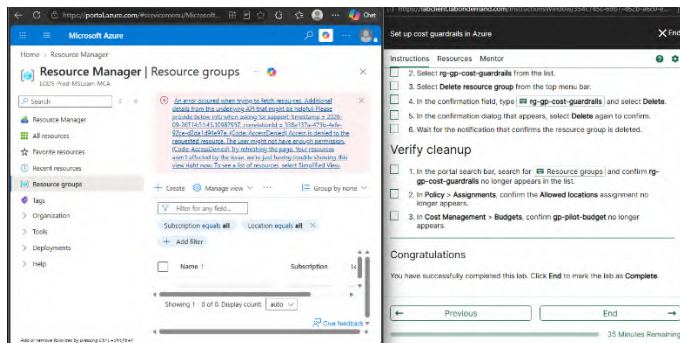
In the provided image, the user is navigating to the Policy Assignments page in the Azure portal to begin removing the policy assignment as part of the project cleanup phase.



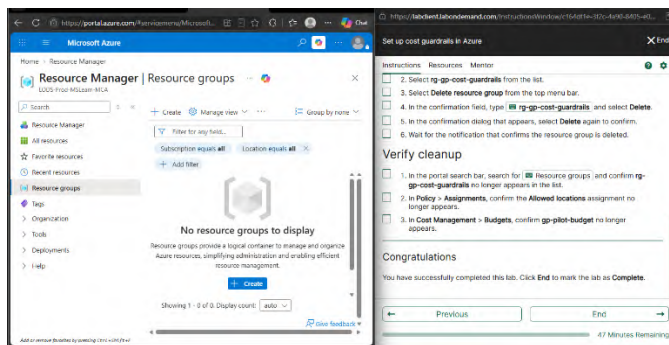
The user successfully deleted the "Allowed locations" policy assignment from the rg-gp-cost-guardrails resource group in the Azure portal as part of the cleanup process.



The user successfully deleted the `rg-gp-cost-guardrails` resource group and all associated resources in the Azure portal to complete the final cleanup phase.

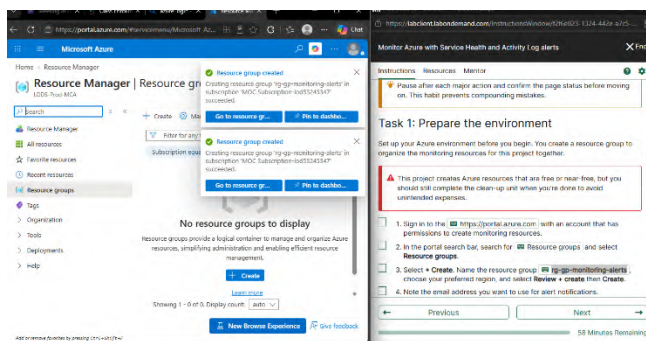


The Azure portal displays an access denied error message while attempting to fetch resources after the `rg-gp-cost-guardrails` resource group has been successfully deleted, confirming the final cleanup and completion of the cloud governance lab.

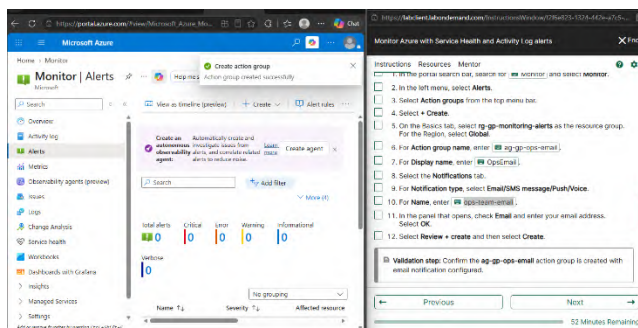


The user verified the final cleanup phase in the Azure portal by confirming that the rg-gp-cost-guardrails resource group no longer appears, successfully completing the cloud governance lab.

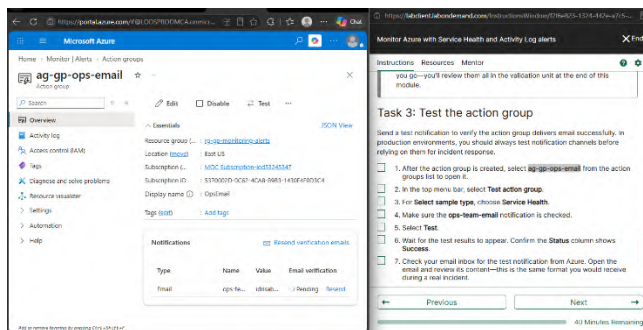
Monitor Azure with Service Health and Activity Log alerts.



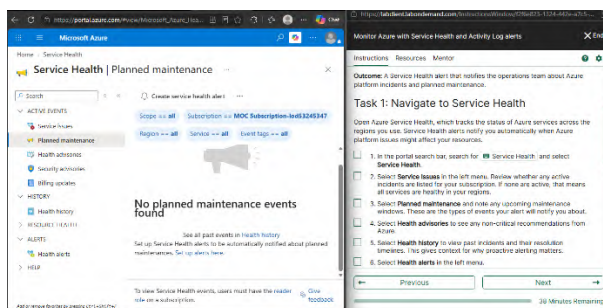
In the provided image, the user successfully created a new resource group named `rg-gp-monitoring-alerts` in the Azure portal to prepare the environment for a new monitoring and activity log alerts lab.



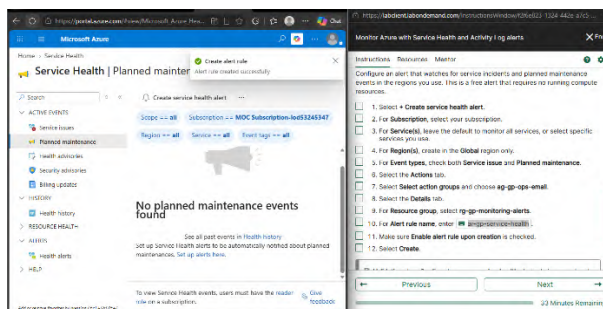
In the provided image, the user successfully created an action group named `ag-gp-ops-email` within the Azure Monitor dashboard to handle email notifications for operational alerts.



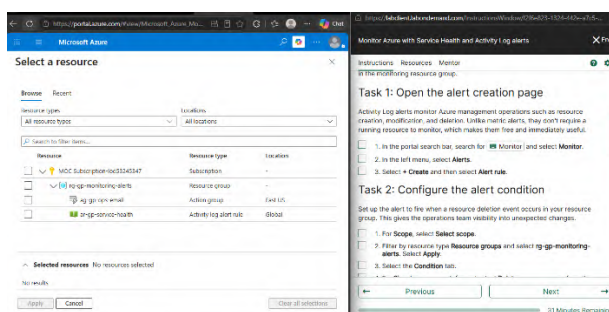
In the provided image, the user is viewing the overview of the `ag-gp-ops-email` action group within the Azure portal to test the action group and verify email notifications.



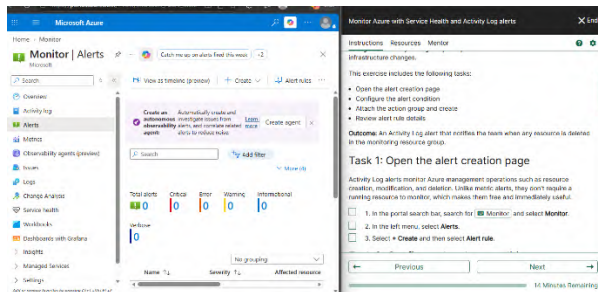
In the provided image, the user is navigating through the Azure Service Health dashboard to review planned maintenance and monitor platform health and advisory status.



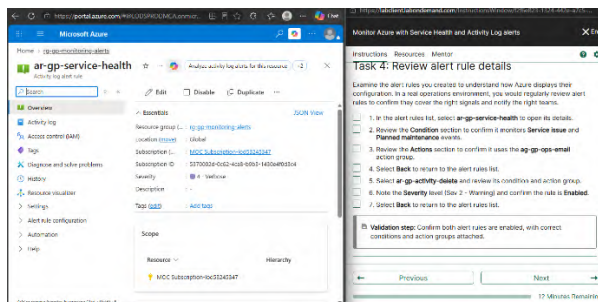
In the provided image, the user successfully created a new service health alert rule named `ar-gp-service-health` within the Azure portal, linking it to the action group to monitor service incidents and planned maintenance.



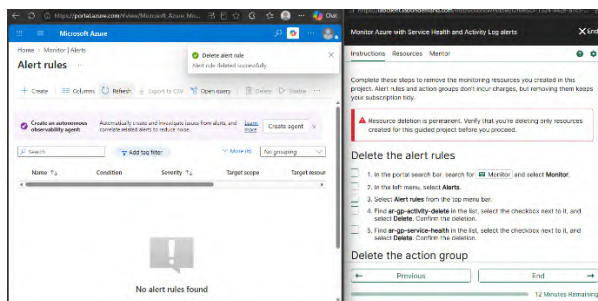
In the provided image, the user is selecting the scope (rg-gp-monitoring-alerts) in the Azure portal to configure an Activity Log alert rule that monitors resource deletion events.



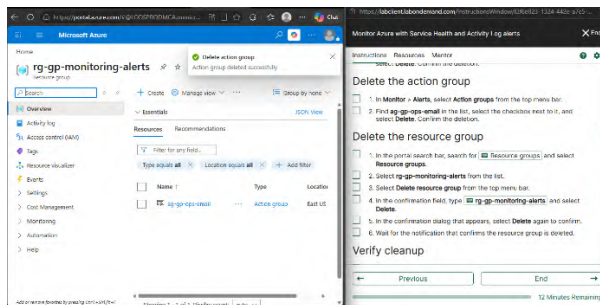
In the provided image, the user is viewing the Azure Monitor Alerts overview page in the portal to begin configuring a new alert rule.



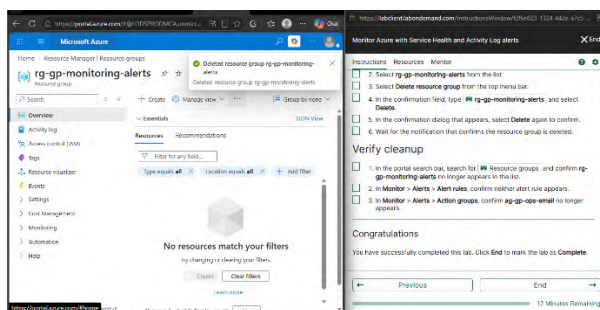
In the provided image, the user is reviewing the configuration details of the `ar-gp-service-health` alert rule within the Azure portal to verify its scope, conditions, and attached action group.



In the provided image, the user successfully deleted the alert rules from the Azure Monitor alerts list in the portal as part of the cleanup phase.

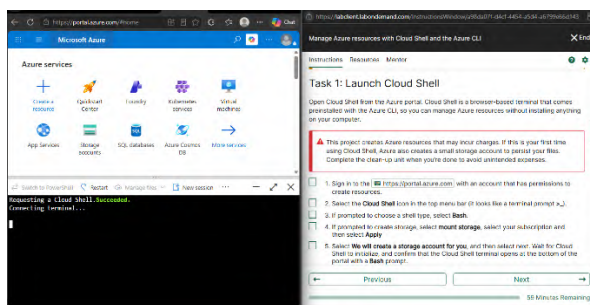


In the provided image, the user successfully deleted the action group `ag-gp-ops-email` from the `rg-gp-monitoring-alerts` resource group in the Azure portal as part of the project cleanup phase.

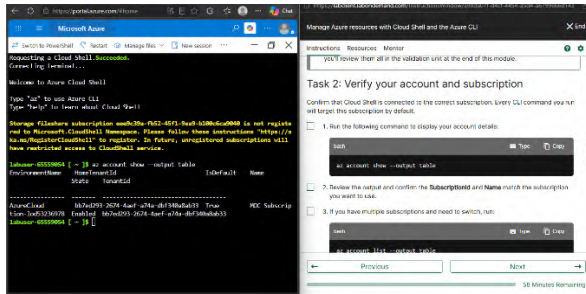


In the provided image, the Azure portal displays a notification confirming the successful deletion of the `rg-gp-monitoring-alerts` resource group, marking the final cleanup and successful completion of the monitoring and alerting lab.

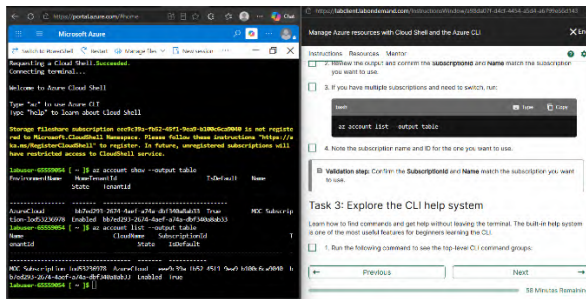
Manage Azure resources with Cloud Shell and the Azure CLI.



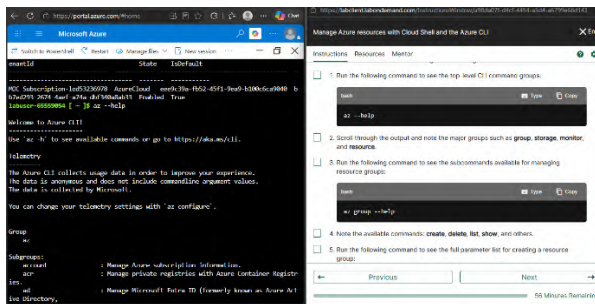
In the provided image, the user is launching the Azure Cloud Shell terminal from the Azure portal to begin managing resources using the Azure CLI.



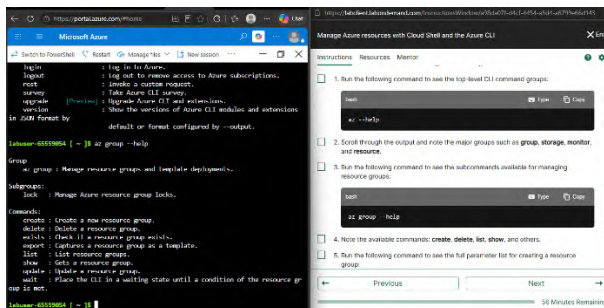
In the provided image, the user executed the ``az account show --output table`` command in Azure Cloud Shell to verify their current groups Azure account details and subscription name.



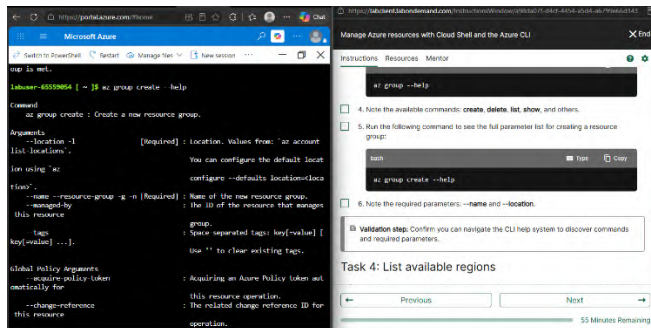
In the provided image, the user executed the ``az account list --output table`` command within Azure Cloud Shell to display and review a table of all accessible Azure subscriptions.



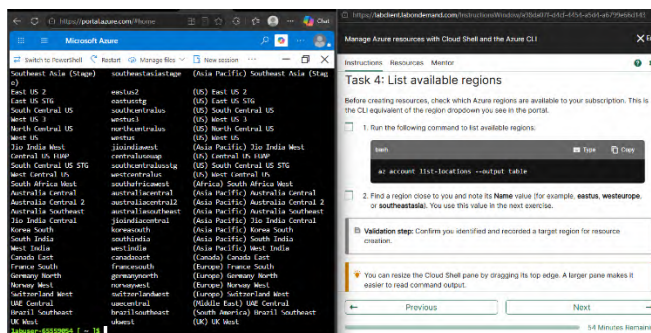
In the provided image, the user executed the ``az --help`` command in Azure Cloud Shell to display top-level command groups and explore the CLI help system.



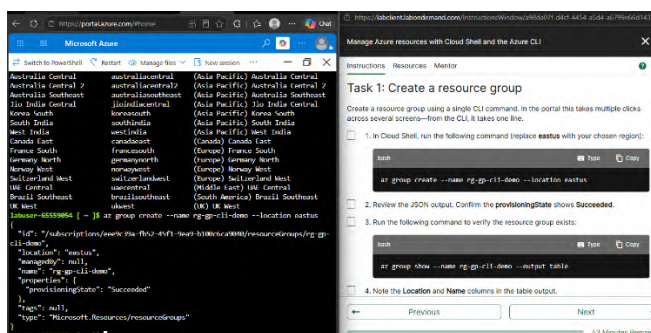
In the provided image, the user executed the `az group --help` command within Azure Cloud Shell to display the subcommands and options available for managing resource groups.



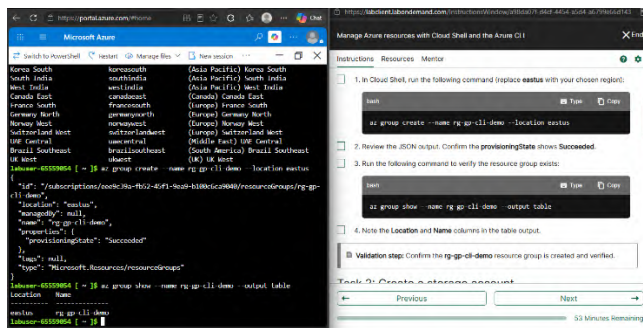
In the provided image, the user executed the `az group create --help` command in Azure Cloud Shell to display the full parameter list and required arguments for creating a resource group.



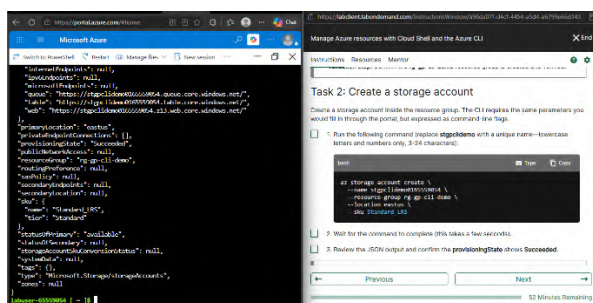
In the provided image, the user executed the `az account list-locations --output table` command within Azure Cloud Shell to display a table of all available Azure regions and their corresponding names for resource deployment.



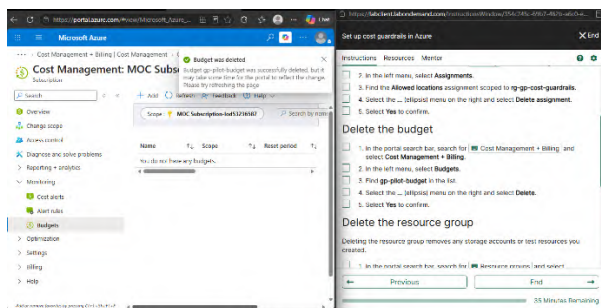
In the provided image, the user executed the `az group create` command within Azure Cloud Shell to successfully provision a new resource group named `rg-gp-cli-demo` in the `eastus` region.



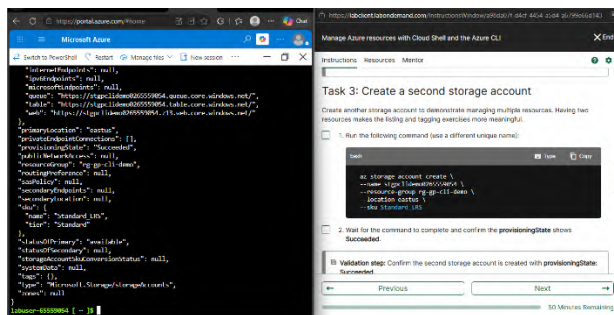
In the provided image, the user executed the `az group show --name rg-gp-cli-demo --output table` command within Azure Cloud Shell to verify the existence of the resource group and display its details in a table format.



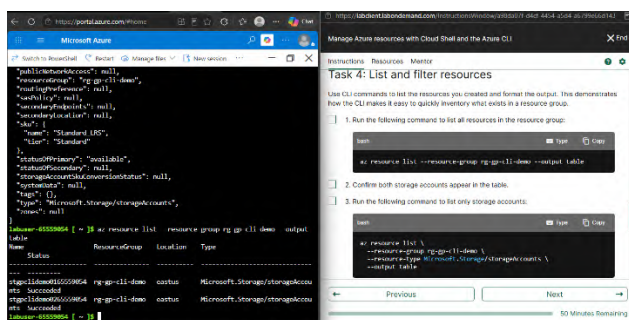
In the provided image, the user executed the `az storage account create` command within Azure Cloud Shell to successfully provision a new storage account named `stgpcldemo0165559054` inside the `rg-gp-cli-demo` resource group.



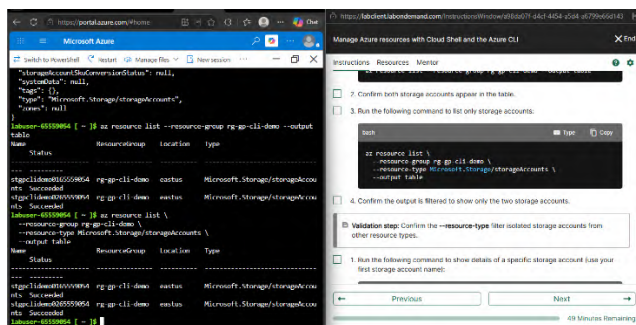
In the provided image, the user successfully deleted the `gp-pilot-budget` budget within the Azure portal's Cost Management section as part of the lab's cleanup process.



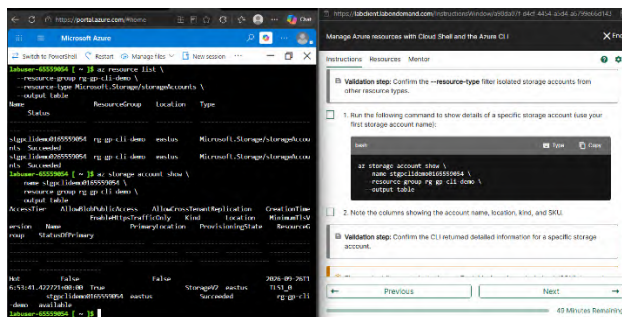
In the provided image, the user executed the ``az storage account create`` command within Azure Cloud Shell to successfully provision a second storage account named ``stgpcldemo0265559054`` inside the ``rg-gp-cli-demo`` resource group.



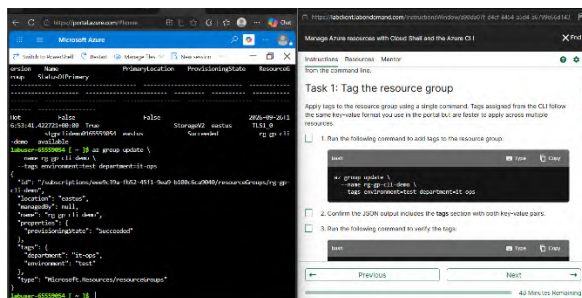
In the provided image, the user executed the ``az resource list`` command within Azure Cloud Shell to display a table of all resources located inside the ``rg-gp-cli-demo`` resource group.



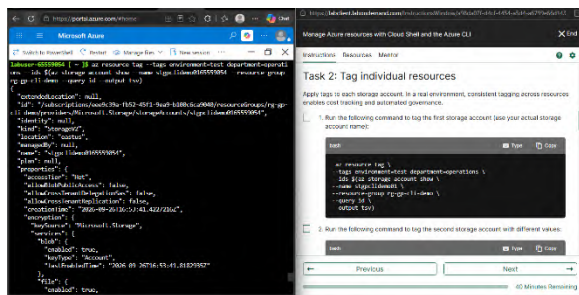
In the provided image, the user executed the filtered resource list command ``az resource list --resource-group rg-gp-cli-demo --resource-type Microsoft.Storage/storageAccounts --output table`` in Azure Cloud Shell to successfully isolate and display only the two storage accounts within the resource group.



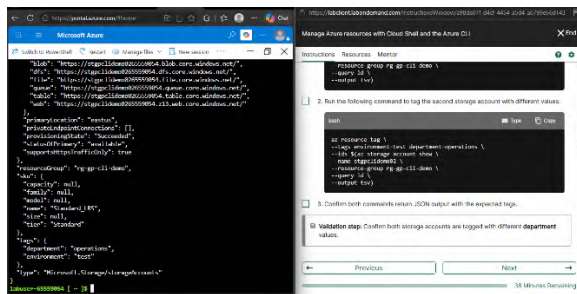
In the provided image, the user executed the `az storage account show` command within Azure Cloud Shell to display detailed properties (such as AccessTier, Kind, Location, and ProvisioningState) of the first storage account (`stgpcldemo0165559054`) in a tabular format.



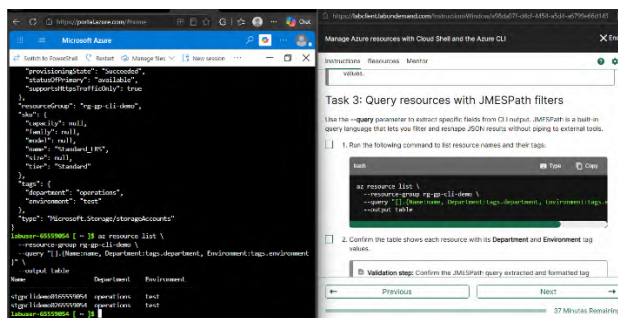
In the provided image, the user executed the `az group update` command within Azure Cloud Shell to successfully apply key-value tags (`environment=test` and `department=it-ops`) to the `rg-gp-cli-demo` resource group.



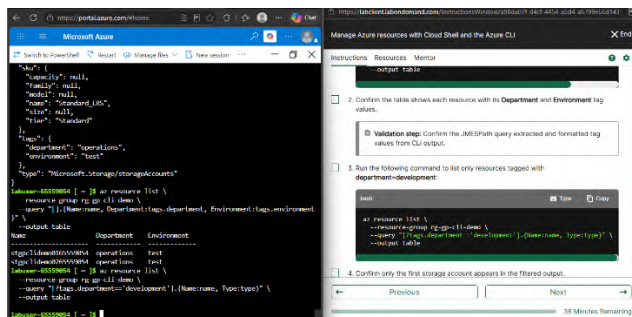
In the provided image, the user executed an Azure CLI command using `az resource tag` combined with a nested `az storage account show` command to apply custom tags (`environment=test` and `department=operations`) to the first storage account (`stgpcldemo0165559054`) within the `rg-gp-cli-demo` resource group.



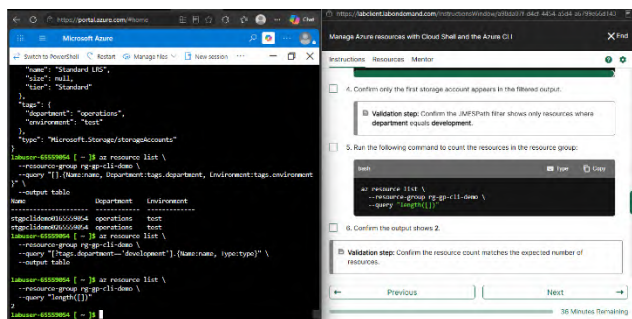
In the provided image, the user executed the command to tag the second storage account (`stgpcldemo0265559054`) with the tags `environment=test` and `department=operations`, successfully confirming the JSON output with the expected tags and completing the task.



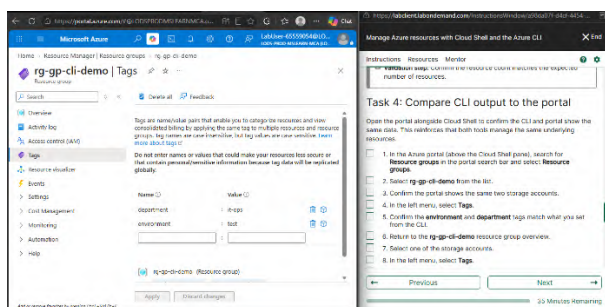
In the provided image, the user executed an `az resource list` command utilizing a JMESPath query within Azure Cloud Shell to extract and format specific resource tags (`Department` and `Environment`) into a clean tabular output for the resources within `rg-gp-cli-demo`.



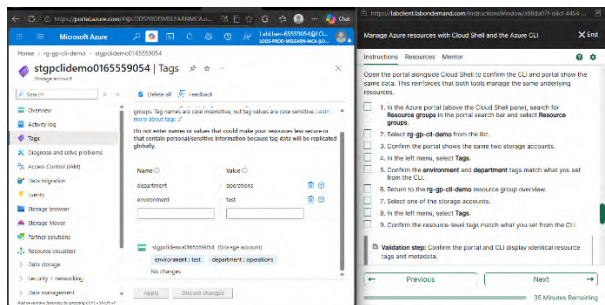
In the provided image, the user executed an `az resource list` command using a JMESPath query filter (`[?tags.department=='development']`) within Azure Cloud Shell to find resources tagged with `department=development` inside the `rg-gp-cli-demo` resource group. Since both storage accounts were tagged with `department=operations`, the output returned an empty table.



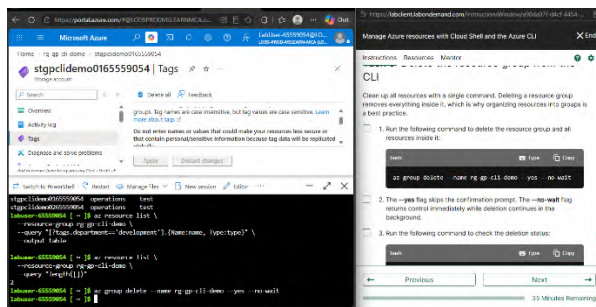
In the provided image, the user successfully executed the `az resource list --resource-group rg-gp-cli-demo --query "length([])"` command in Azure Cloud Shell, and the JMESPath query correctly returned a total count of 2 resources.



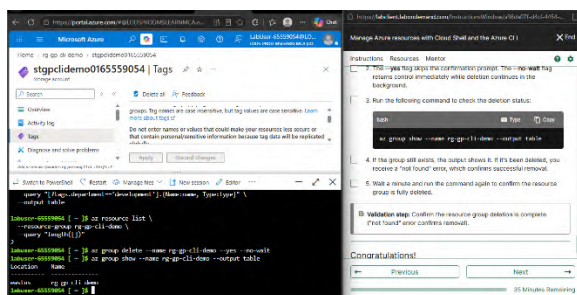
In the provided image, the user is viewing the **Tags** configuration blade for the `rg-gp-cli-demo` resource group in the Azure portal, confirming that the tags (`department: it-ops` and `environment: test`) applied via the Azure CLI match what is shown in the portal.



In the provided image, the user is viewing the Tags configuration blade for the first storage account (`stgpcldemo0165559054`) in the Azure portal, verifying that the resource-level tags (`department: operations` and `environment: test`) match what was previously configured via the Azure CLI.

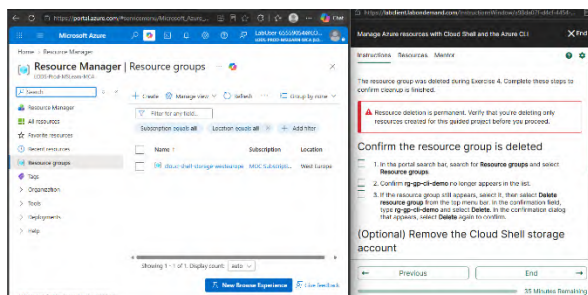


In the provided image, the user executed the `az group delete --name rg-gp-cli-demo --yes --no-wait` command within Azure Cloud Shell to initiate the asynchronous deletion of the resource group and all of its contained resources without requiring a confirmation prompt.



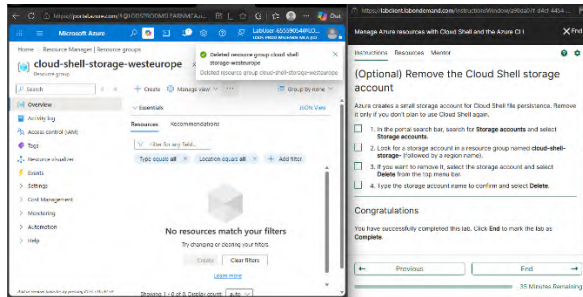
In the provided image, the user executed the `az group show --name rg-gp-cli-demo --output table` command in Azure Cloud Shell to check the deletion status of the resource group. Since the output still displays the resource group `rg-gp-cli-demo` in the `eastus` location, the background deletion is still in progress.

To complete the lab verification, you can wait a moment and run the command again until it returns a "not found" error, confirming that the resource group and all its contained resources have been successfully removed.



In the provided image, the Azure portal's Resource groups view confirms that `rg-gp-cli-demo` no longer appears in the list, verifying that the resource group and all of its contained resources have been successfully deleted. Only the default `cloud-shell-storage-westeurope` storage account remains.

You have successfully completed all the steps for resource management, tagging, JMESPath querying, and cleanup in this Azure CLI lab! Let me know if you are ready to wrap up or move on to the next task.

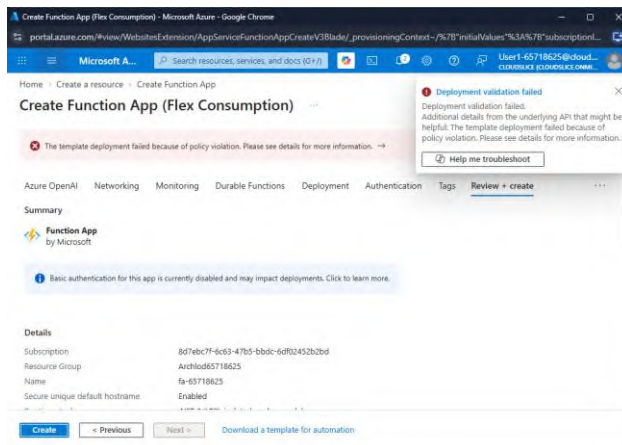


In the provided image, the Azure portal shows a success notification confirming that the optional Cloud Shell storage account resource group (cloud-shell-storage-westeurope) has been deleted. The instructions on the right indicate that you have successfully completed the entire lab.

You are ready to click End in the lab window to mark your progress as complete! Let me know if you would like to review anything else from your Azure CLI exercises.

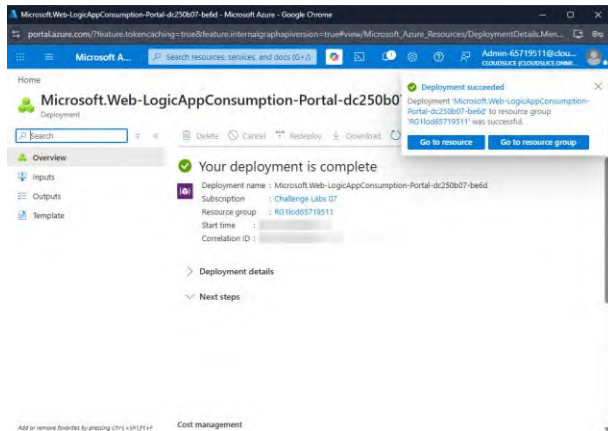
Skillable Challenges supporting AZ-900: Describe cloud concepts

Create an Azure Function App [Guided]

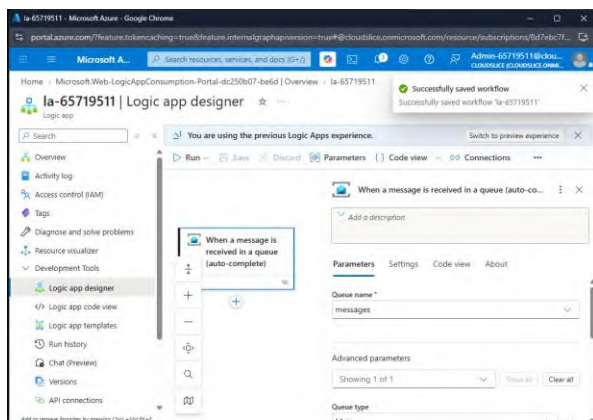


The user attempted to review and create a Flex Consumption Function App in the Azure portal, but the deployment validation failed due to an active policy violation.

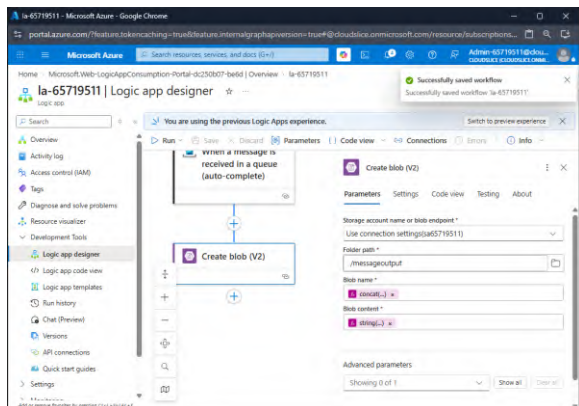
Create an Azure Logic App [Guided]



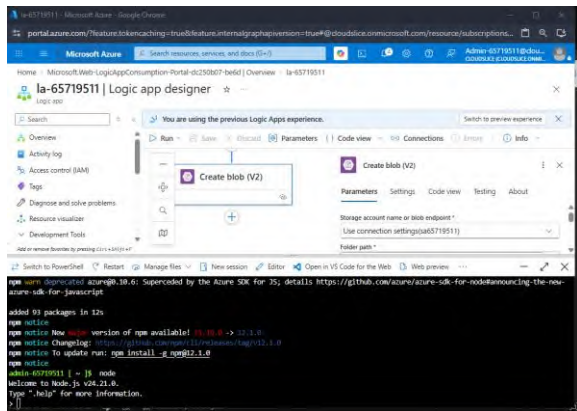
The Azure portal displays a success message confirming that the deployment of the Logic App (Microsoft.Web-LogicAppConsumption-Portal-dc250b07-be6d) to the resource group RG1lod65719511 has completed successfully.



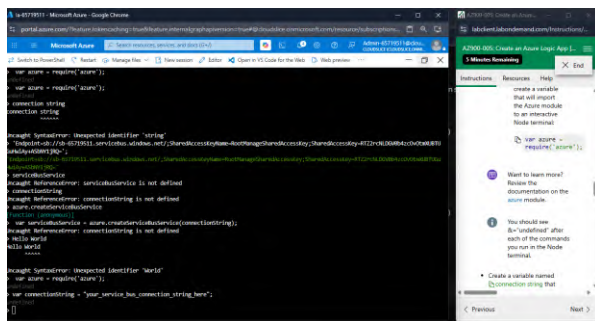
The user successfully saved the Logic App workflow configuration for the trigger "When a message is received in a queue (auto-complete)" in the Azure portal.



The user successfully configured the parameters for the "Create blob (V2)" action step within the Logic App designer interface in the Azure portal.

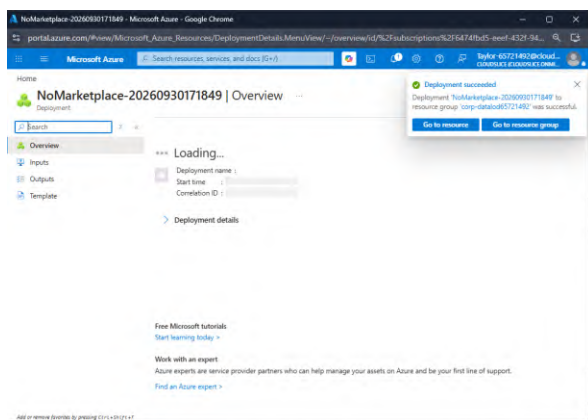


The user successfully installed the deprecated `azure` package via npm and launched an interactive Node.js terminal session in Azure Cloud Shell while configuring a Logic App.

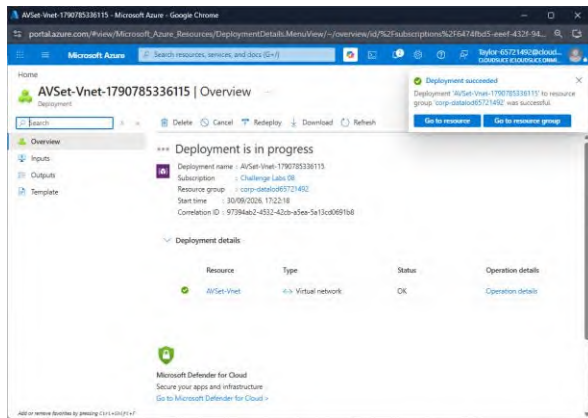


The user tested and debugged Node.js terminal commands in Azure Cloud Shell to import the Azure module and initialize variables for the Service Bus connection.

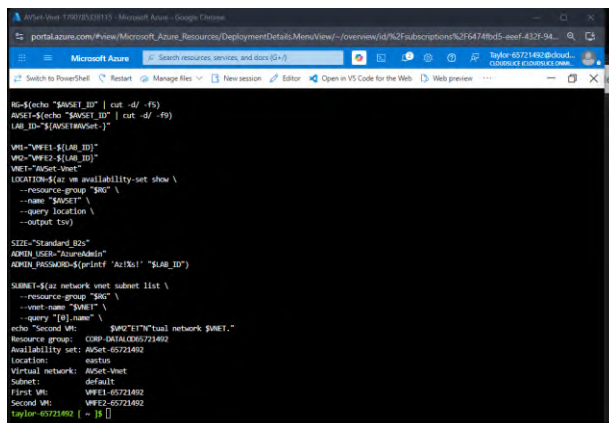
Enable High Availability by Using Availability Sets [Guided]



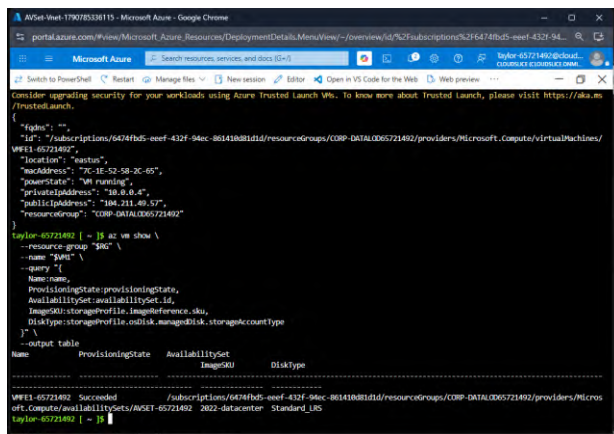
The Azure portal displays a success notification confirming that the deployment (``NoMarketplace-20260930171849``) to the resource group ``corp-datalod65721492`` has successfully completed.



The Azure portal displays a success notification confirming that the deployment of the virtual network (``AVSet-Vnet-1790785336115``) to the resource group ``corp-datalod65721492`` has successfully completed.



The user executed a bash script in Azure Cloud Shell to set environment variables and retrieve resource details for an availability set and virtual machines.



```
Switch to PowerShell | Restart | Manage files | New session | Editor | Open in VS Code for the Web | Web preview | ...

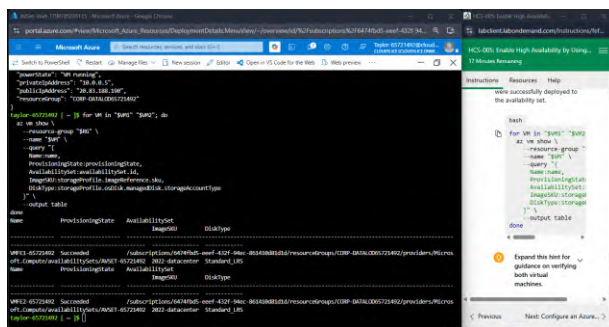
Consider upgrading security for your workloads using Azure Trusted Launch VMs. To know more about Trusted Launch, please visit https://aka.ms/TrustedLaunch.

{
  "fqdns": "",
  "id": "/subscriptions/6474bdc5-eeef-432f-94ec-861410b81d1d/resourceGroups/CRP-DATA/005721492/providers/Microsoft.Compute/virtualMachines/VMFE1-65721492",
  "location": "eastus",
  "macAddress": "7C-1E-52-58-2C-85",
  "powerState": "VM running",
  "privateIpAddress": "10.0.0.4",
  "publicIpAddress": "104.211.40.57",
  "resourceGroup": "CRP-DATA/005721492"
}

taylor@65721492: ~$ az vm show \
  --resource-group "jms" \
  --name "jms" \
  --query '[Name,ProvisioningState,AvailabilitySet,ImageSKU,DiskType]' \
  --output table

Name      ProvisioningState  AvailabilitySet  ImageSKU  DiskType
-----
VMFE1-65721492  Succeeded         /subscriptions/6474bdc5-eeef-432f-94ec-861410b81d1d/resourceGroups/CRP-DATA/005721492/providers/Microsoft.Compute/availabilitysets/AVSET-65721492-2022-datascanner  Standard_LRS
taylor@65721492: ~$
```

The user executed an Azure CLI command in Cloud Shell to query and display specific properties—including provisioning state, availability set, image SKU, and disk type—for the virtual machine `VMFE1-65721492`.



```
port.azure.com/View/Microsoft_Azure_Resource/DeploymentDetails/MenuView/overview/%2Fsubscriptions%2F6474bdc5-eeef-432f-94ec-861410b81d1d/resourceGroups/CRP-DATA/005721492/providers/Microsoft.Compute/virtualMachines/VMFE1-65721492

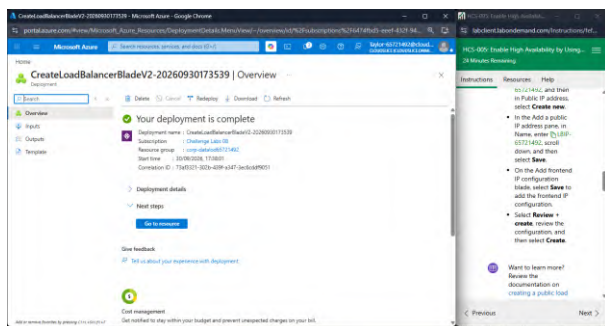
Switch to PowerShell | Restart | Manage files | New session | Editor | Open in VS Code for the Web | Web preview | ...

"powerState": "VM running",
"privateIpAddress": "10.0.0.4",
"publicIpAddress": "104.211.40.57",
"resourceGroup": "CRP-DATA/005721492"
}

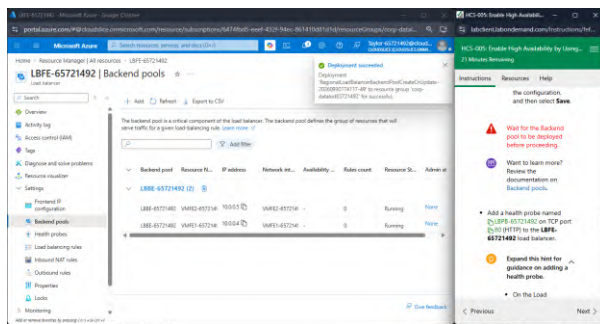
taylor@65721492: ~$ for VM in $(az vm show \
  --resource-group "jms" \
  --name "jms" \
  --query '[Name,ProvisioningState,AvailabilitySet,ImageSKU,DiskType]' \
  --output table
do
  az vm show \
  --resource-group "jms" \
  --name "jms" \
  --query '[Name,ProvisioningState,AvailabilitySet,ImageSKU,DiskType]' \
  --output table
done

Name      ProvisioningState  AvailabilitySet  ImageSKU  DiskType
-----
VMFE1-65721492  Succeeded         /subscriptions/6474bdc5-eeef-432f-94ec-861410b81d1d/resourceGroups/CRP-DATA/005721492/providers/Microsoft.Compute/availabilitysets/AVSET-65721492-2022-datascanner  Standard_LRS
taylor@65721492: ~$
```

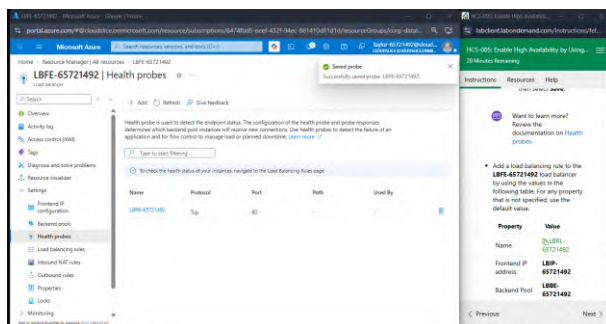
The user ran a bash loop in Azure Cloud Shell using the Azure CLI to query and verify the configuration properties of both virtual machines in the availability set.



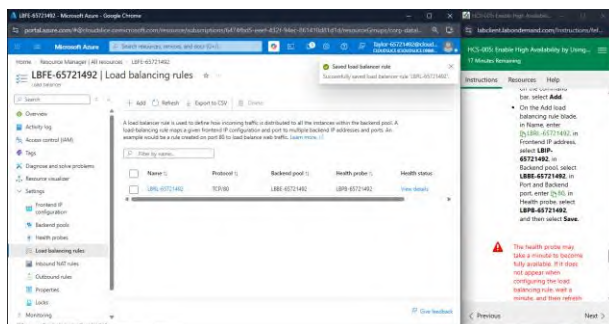
The Azure portal displays a success message confirming that the deployment of the load balancer has been successfully completed.



The Azure portal displays a success notification confirming that the backend pool configuration for the load balancer (`LBFB-65721492`) has successfully completed and populated both virtual machines.

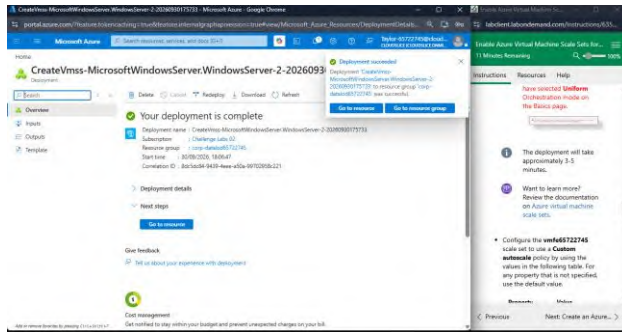


The user successfully saved the health probe configuration (`LBPB-65721492`) for the load balancer in the Azure portal.

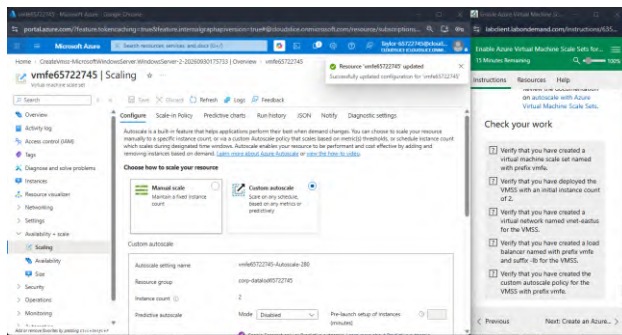


The user successfully created and saved the load balancing rule (`LBRL-65721492`) for the load balancer in the Azure portal.

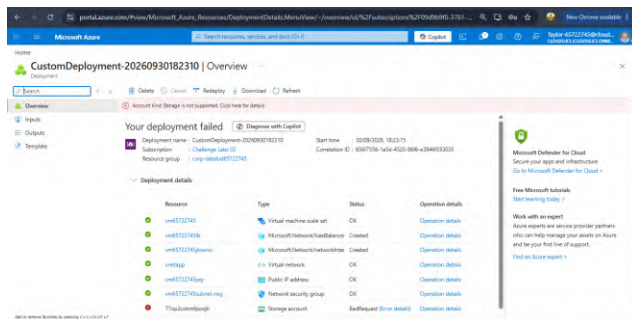
Enable Azure Virtual Machine Scale Sets for High Availability and Scalability [Guided]



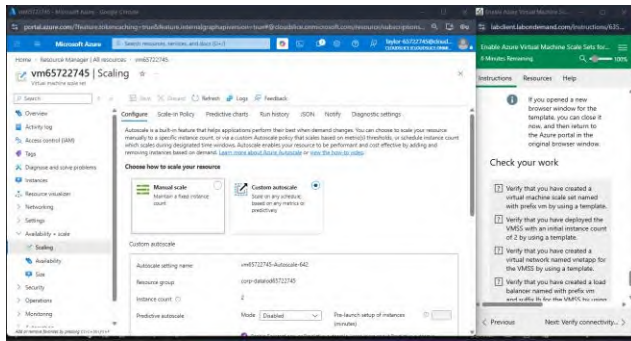
The Azure portal displays a success message confirming that the deployment of the Windows Server Virtual Machine Scale Set to the resource group `corp-data65722745` has been successfully completed.



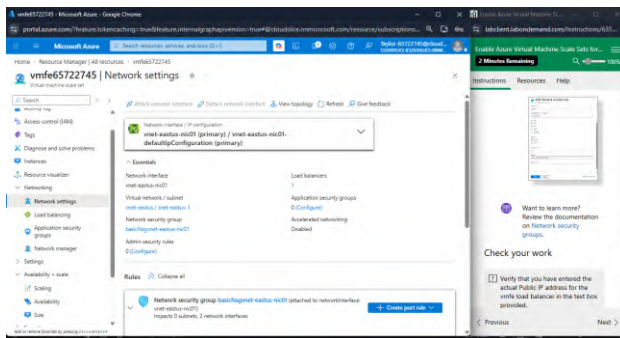
The user successfully updated and saved the custom autoscale policy configuration for the Virtual Machine Scale Set (`vmfe65722745`) in the Azure portal.



The Azure portal deployment view indicates that the custom template deployment (`CustomDeployment-20260930182310`) failed due to an unsupported account kind error on the storage account resource, while other components like the virtual machine scale set and virtual network were successfully created.



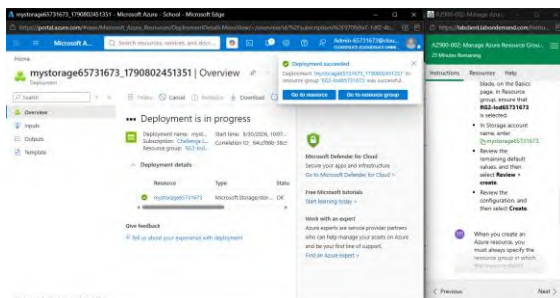
The user successfully verified the custom autoscale and scaling configuration for the Virtual Machine Scale Set (`vm65722745`) in the Azure portal while following the lab check-your-work instructions.



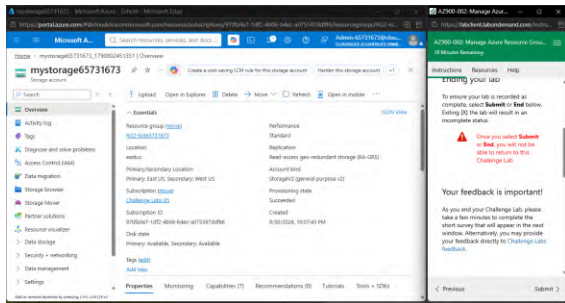
The user viewed the network settings and configuration details for the Virtual Machine Scale Set (`vmfe65722745`) in the Azure portal to verify its network interface, virtual network, and associated security rules.

Skillable Challenges supporting AZ-900: Describe Azure architecture and services

Manage Azure Resource Groups [Guided]

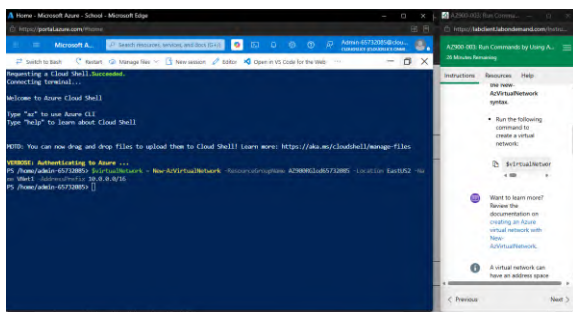


The Azure portal displays a success notification confirming that the deployment of the storage account (`mystorage65731673`) to the resource group `RG2-lod65731673` has successfully completed.

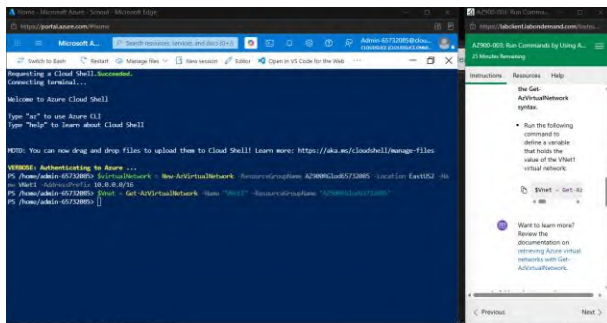


The user viewed the overview and properties page of the successfully deployed storage account (` mystorage65731673`) in the Azure portal while finishing up the lab instructions.

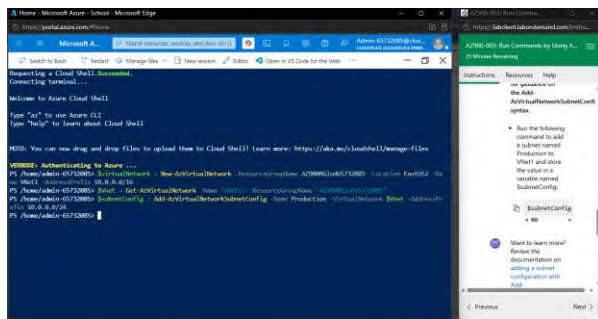
Run Commands by Using Azure Cloud Shell [Guided]



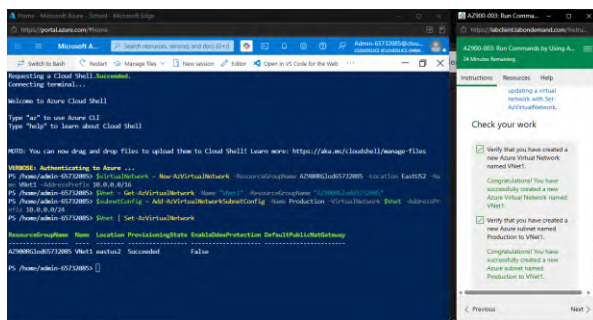
The user initialized Azure Cloud Shell in PowerShell and executed the `New-AzVirtualNetwork` cmdlet to create a new virtual network within the specified resource group.



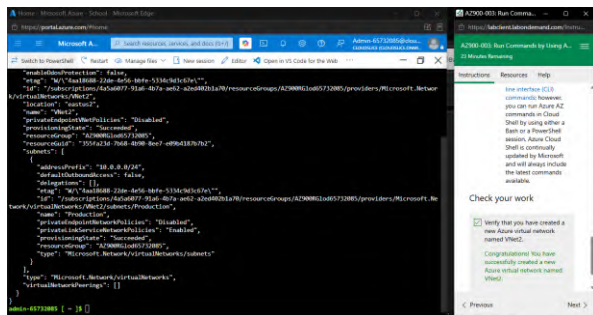
The user executed the `Get-AzVirtualNetwork` cmdlet in Azure PowerShell to retrieve and store the configuration of the virtual network `VNet1` in a variable.



The user executed the `Add-AzVirtualNetworkSubnetConfig` cmdlet in Azure PowerShell to define a new subnet named `Production` with the address prefix `10.0.0.0/24` and store the configuration in the `\$subnetConfig` variable.

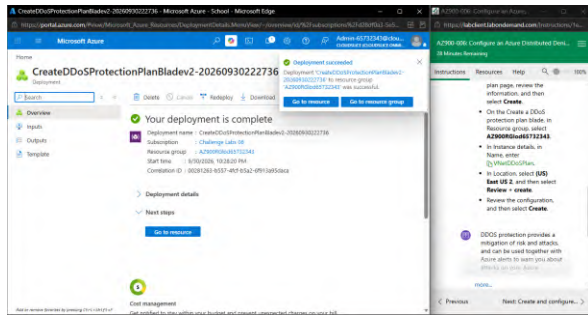


The user executed the `Set-AzVirtualNetwork` cmdlet in Azure PowerShell to apply the updated subnet configuration to `VNet1` and successfully completed the verification checks in the lab instructions.

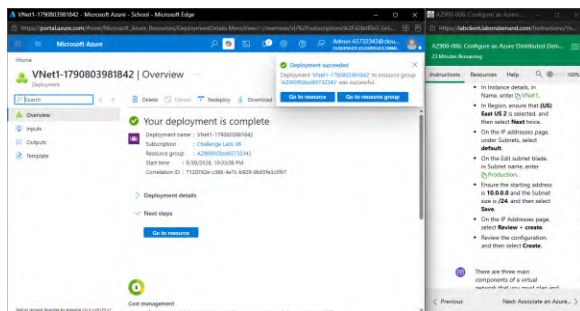


The user ran Azure CLI commands in Cloud Shell to verify the successful creation of the virtual network `VNet2` alongside its configured subnet details.

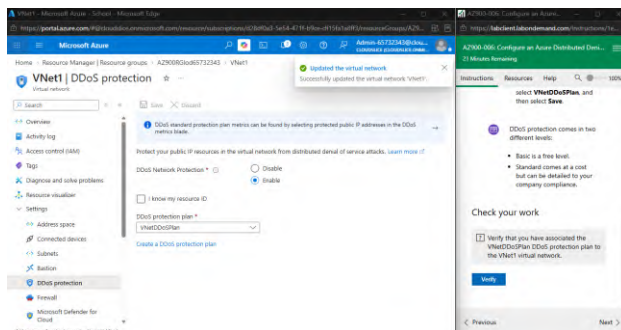
Configure an Azure Distributed Denial of Service Protection Plan [Guided]



The Azure portal displays a success notification confirming that the deployment of the DDoS protection plan (`VNetDDoSPlan`) to the resource group `AZ900RGlod65732343` has successfully completed.



The Azure portal displays a success notification confirming that the deployment of the virtual network (`VNet1`) to the resource group `AZ900RGlod65732343` has successfully completed.



The Azure portal displays a success notification confirming that the deployment of the virtual network (`VNet1`) to the resource group `AZ900RGlod65732343` has successfully completed.

The screenshot shows the Microsoft KeyVault 'Your deployment is complete' page. The page displays the deployment details for 'keyVault56732670', including the deployment name, change log, resource group, start time, and completion time. The deployment is marked as 'Succeeded'. The page also provides instructions for the next steps, including enabling volume encryption, reviewing the configuration, and creating a new deployment.

Deployment Details:

- Deployment name: keyVault56732670
- Change log: 1
- Resource group: A7290-001-15070-001-sub001
- Start time: 8/3/2024, 10:40:28 PM
- Completion time: 8/3/2024, 11:16:26 AM (UTC+05:30)

Next Steps:

- On the Networking page, in Enable profile access, select the check box, and then select **Next**.
- On the Networking page, in Enable profile access, select the check box, and then select **Next**.
- Review the remaining values, and then select **Review + create**.
- Review the configuration, and then select **Create**.

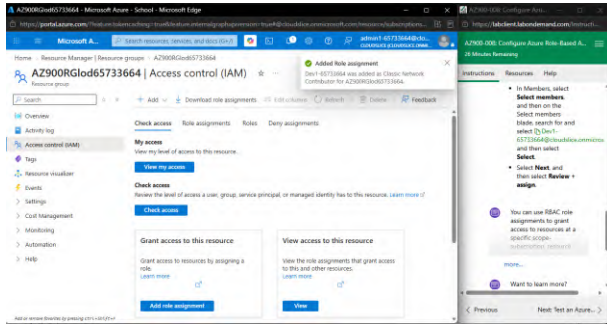
The screenshot shows the Microsoft Azure portal interface. The main heading is 'KeyVault65732670 | Keys'. Below this, a message states: 'The key Salkey-65732670 has been successfully created.' A table lists the key with the following details:

Name	Status	Expiration date
Salkey-65732670	Enabled	

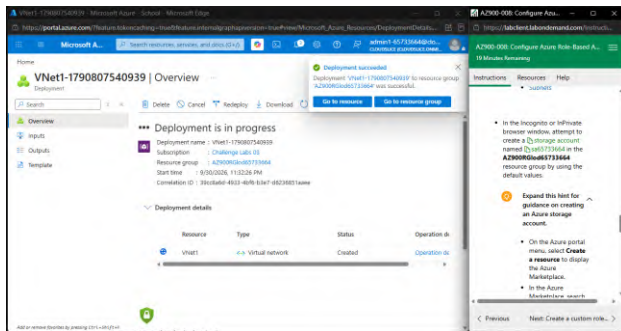
The left-hand navigation pane includes links to Overview, Activity log, Access control (IAM), Keys (selected), Diagnostics and issue problems, Access policies, Resource visualizer, Events, Objects, and a sub-menu for Keys containing Secrets, Certificates, Settings, and Monitoring. The right-hand pane, titled 'Instructions', provides a list of steps to create a key, including selecting the KeyVault, clicking 'Create a key', choosing 'Salkey-65732670' as the name, and clicking 'Create'.

[illegible]

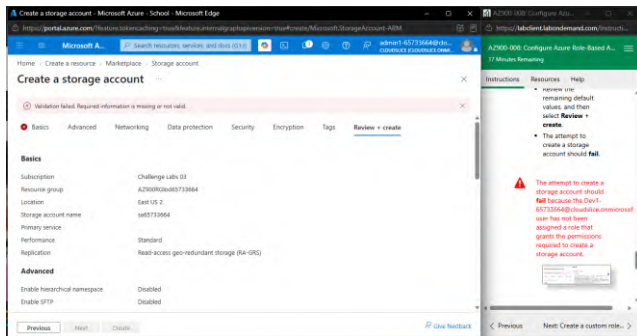
Configure Azure Role-Based Access Control [Guided]



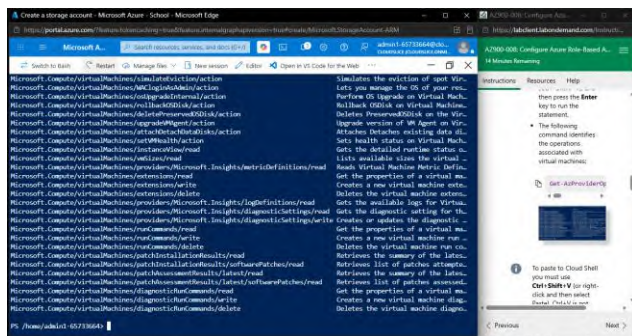
The user successfully configured and updated the storage account (`sa65732670`) to use customer-managed encryption keys from the Key Vault, as confirmed by the success notification in the Azure portal.



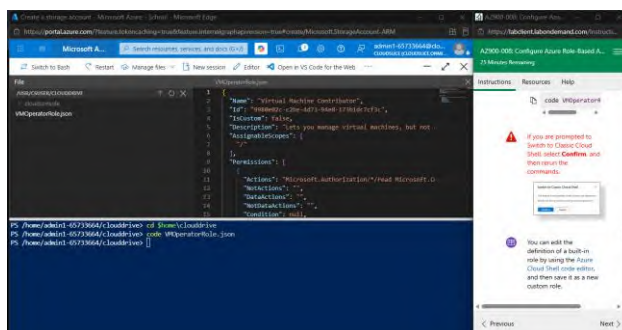
The user successfully configured and updated the storage account (sa65732670) to use customer-managed encryption keys from the Key Vault, as confirmed by the success notification in the Azure portal.



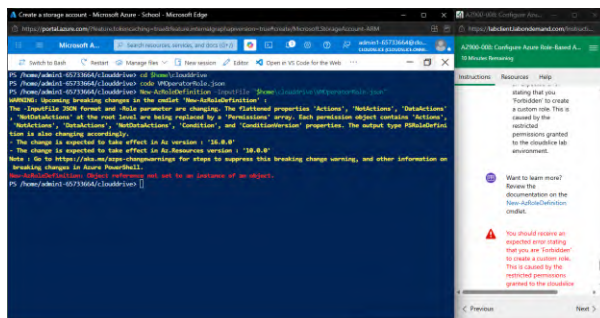
The Azure portal shows a validation failure while attempting to create the storage account (`sa65733664`) because the user lacks the necessary permissions required for the action.



The user executed the `Get-AzProviderOperation` cmdlet in the Azure Cloud Shell to display the operations and permissions associated with virtual machines for custom role configuration.

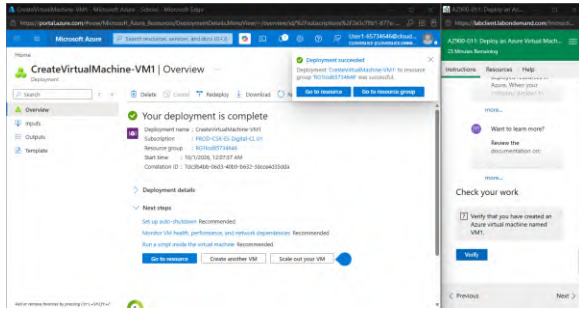


The user opened the Cloud Shell code editor to create and edit a JSON file (`VMOperatorRole.json`) for defining a custom role configuration in Azure.



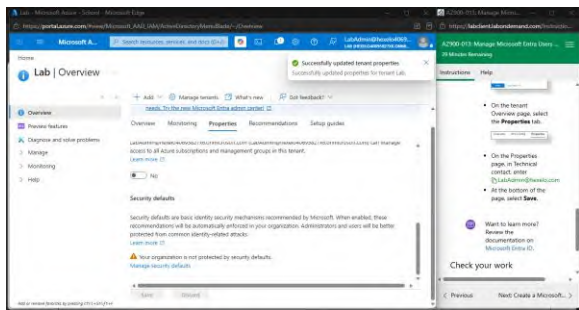
The user executed the `New-AzRoleDefinition` cmdlet in PowerShell using the JSON configuration file, resulting in an error due to the restricted permissions in the lab environment.

Deploy an Azure Virtual Machine [Guided]

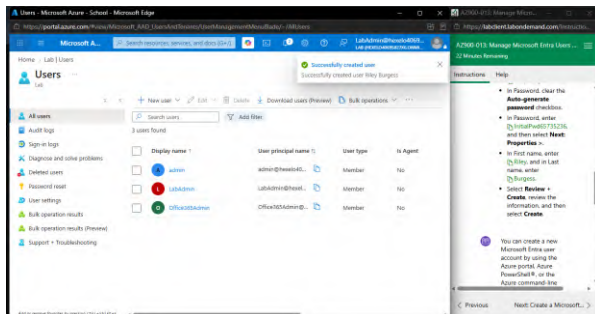


The Azure portal displays a success notification confirming that the deployment of the virtual machine (`VM1`) to the resource group `RG1lod65734646` has successfully completed.

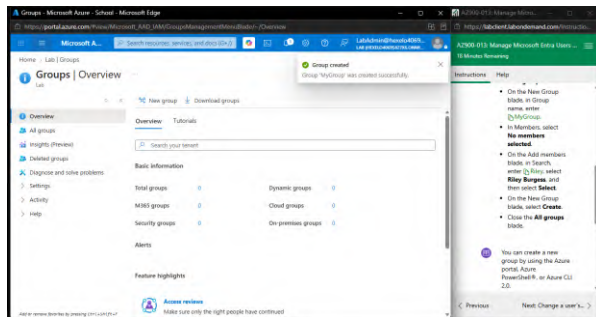
Manage Microsoft Entra Users and Groups [Guided]



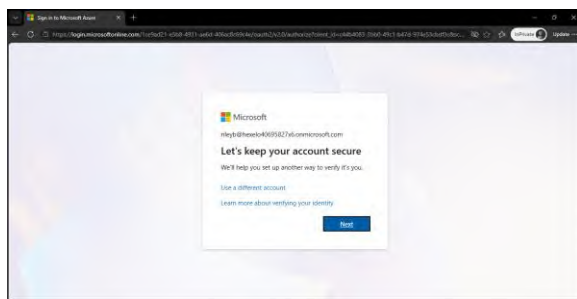
The user successfully updated the tenant properties for the Microsoft Entra ID organization in the Azure portal, as confirmed by the success notification.



The user successfully created a new Microsoft Entra user account for Riley Burgess within the Azure portal, as confirmed by the success notification.

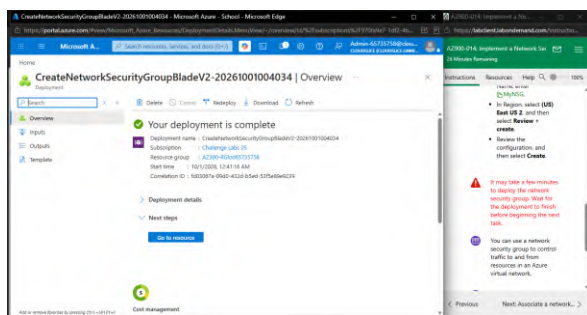


The user successfully created a new Microsoft Entra security group named MyGroup and assigned the newly created user Riley Burgess as a member, as confirmed by the notification in the Azure portal.

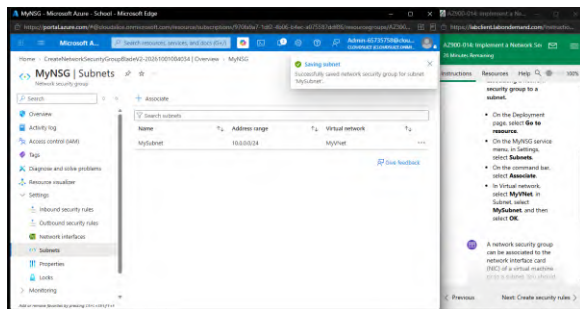


The user is signing in with the newly created user account (`rileyb@hexelo40695827x6.onmicrosoft.com`) in an InPrivate browser window to configure account security verification prompts.

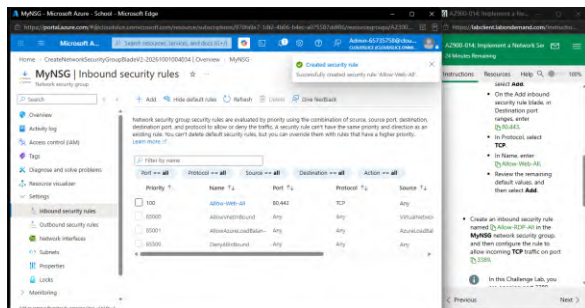
Implement a Network Security Group [Guided]



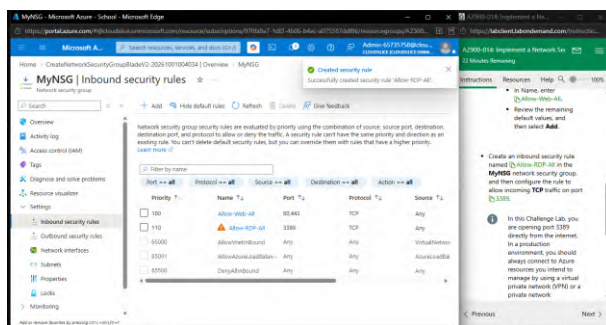
The Azure portal displays a success notification confirming that the deployment of the Network Security Group to the resource group AZ900-RGlod65735758 has successfully completed.



The user successfully associated the network security group (`MyNSG`) with the subnet `MySubnet` in the virtual network `MyVNet` via the Azure portal.



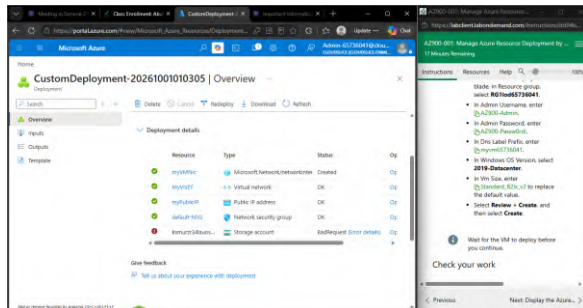
The user successfully created a new inbound security rule named Allow-Web-All in the network security group (MyNSG) to allow incoming TCP traffic on port 80,443 via the Azure portal.



The user successfully created a new inbound security rule named Allow-RDP-All in the network security group (MyNSG) to allow incoming TCP traffic on port 3389 via the Azure portal.

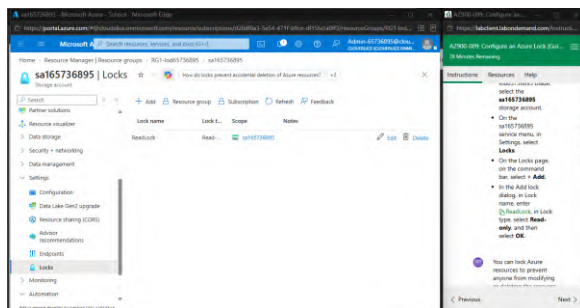
Skillable Challenges supporting AZ-900: Describe Azure management and governance

Manage Azure Resource Deployment by Using an Azure Resource Manager Template [Guided]

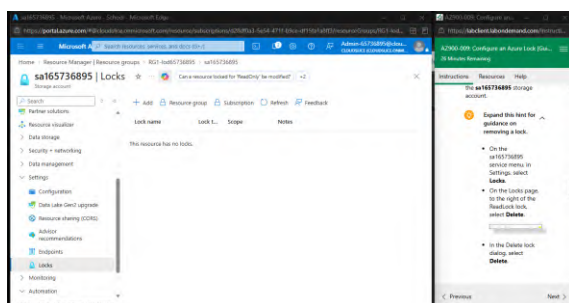


The Azure portal displays a custom template deployment overview (`CustomDeployment-20261001010305`) showing successful provisioning of core networking components like `MyVNET` and `myVNet`, alongside a BadRequest error encountered while provisioning a storage account.

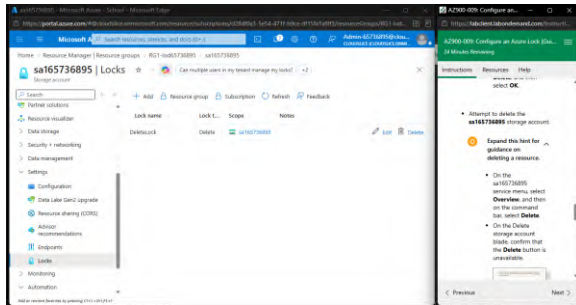
Configure an Azure Lock [Guided]



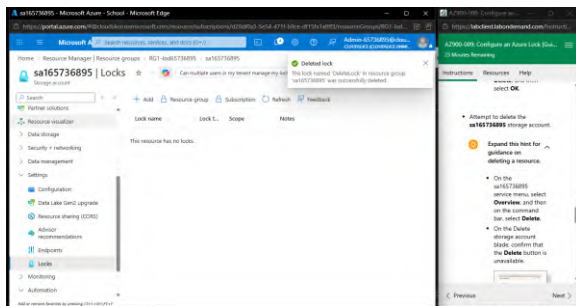
The user successfully added a Read-only resource lock named ReadLock to the storage account (sa165736895) via the Azure portal settings.



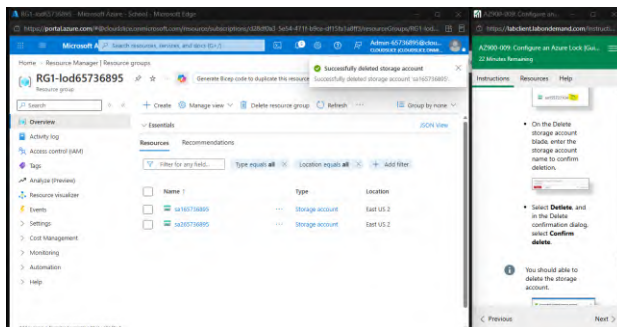
The user successfully deleted the resource lock from the storage account (sa165736895), as shown by the empty locks list in the Azure portal settings.



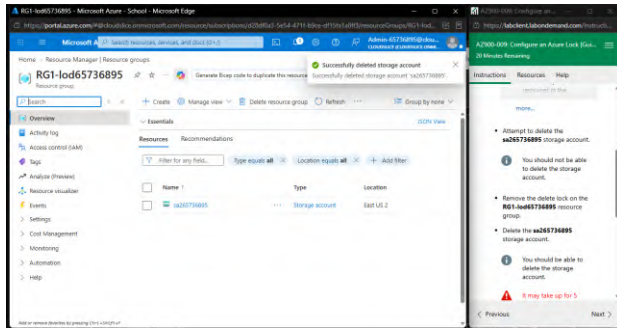
The user successfully added a Delete resource lock named DeletesLock to the storage account (sa165736895) via the Azure portal settings to prevent unauthorized resource deletion.



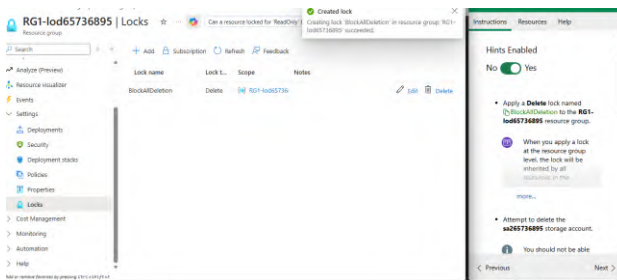
The user successfully deleted the delete lock named `DeletesLock` from the storage account (`sa165736895`) via the Azure portal settings, as confirmed by the success notification.



The Azure portal displays a success notification confirming the successful deletion of the storage account (sa165736895) from the resource group RG1-lod65736895.

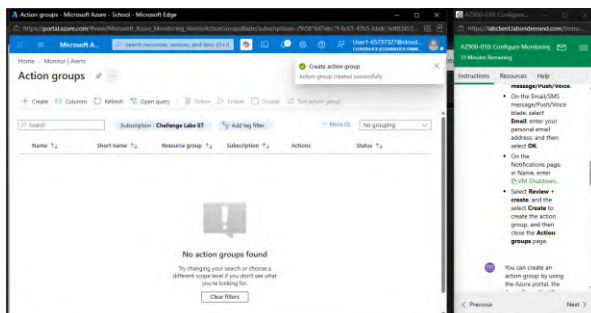


The Azure portal displays a success notification confirming the successful deletion of the second storage account (sa265736895) from the resource group RG1-lod65736895.

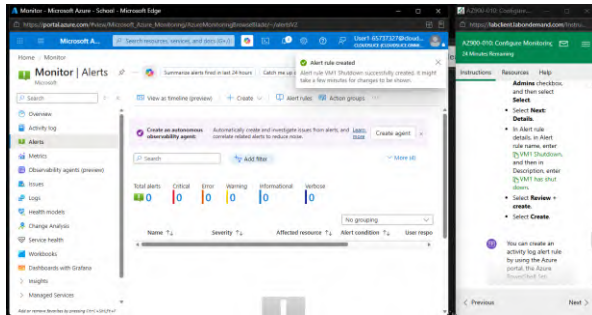


The user successfully applied a Delete resource lock named BlockAllDeletion to the resource group RG1-lod65736895 via the Azure portal, as confirmed by the success notification.

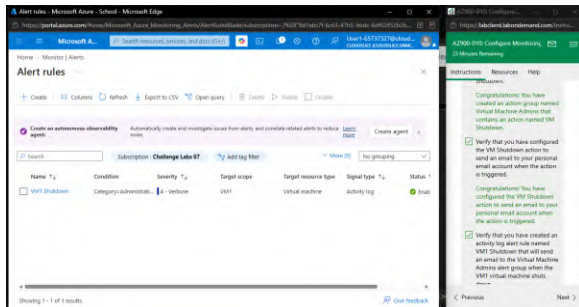
Configure Monitoring by Using Azure Monitor [Guided]



The user successfully created a new Azure Monitor action group via the Azure portal, as confirmed by the success notification.

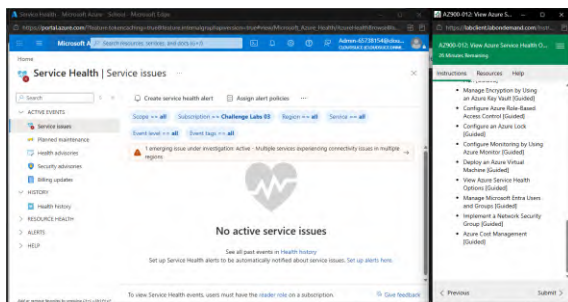


The user successfully created a new Azure Monitor activity log alert rule named VM1 Shutdown via the Azure portal, as confirmed by the success notification.



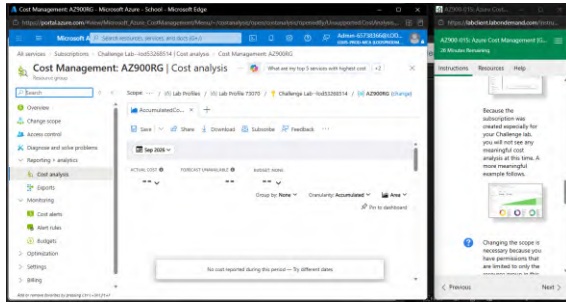
The Azure portal displays the configured alert rules blade listing the active VM1 Shutdown activity log alert rule targeting VM1, alongside verification checkmarks in the lab instructions.

View Azure Service Health Options [Guided]

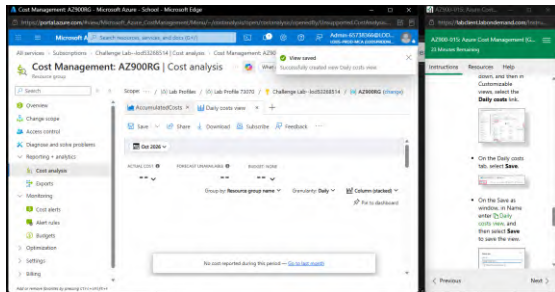


The user accessed the Azure Service Health dashboard in the Azure portal to review service issues, emerging alerts, and regional status across their subscriptions.

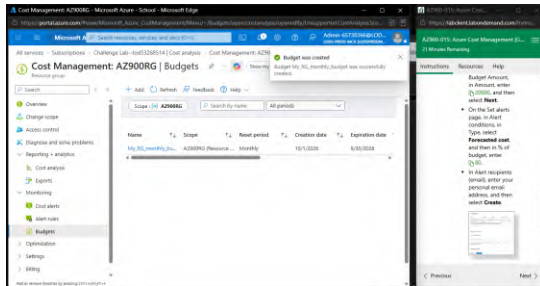
Azure Cost Management [Guided]



The user navigated to the Cost Management and Billing dashboard within the Azure portal to review the Cost Analysis metrics and expenditure reports for the resource group AZ900RG.



The user successfully saved a custom cost analysis configuration named Daily costs view in the Azure Cost Management portal, as confirmed by the success notification.



The user successfully created a new monthly budget named My_RG_monthly_budget for the AZ900RG resource group within the Azure Cost Management portal, as confirmed by the success notification.

End Of Class Survey